

IS THE HOT HAND REAL? EVIDENCE FROM A PERMUTATION AND
HIERARCHICAL-BASED ANALYSIS

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9 Cameron An
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Cameron An

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TITLE: Is the Hot Hand Real? Evidence from a Permutation and
Hierarchical-Based Analysis

AUTHOR: Cameron An

DATE SUBMITTED: June 2026

COMMITTEE CHAIR: Kevin Ross, Ph.D.
Professor of Statistics

COMMITTEE MEMBER: Bret Holladay, Ph.D.
Assistant Professor of Statistics

COMMITTEE MEMBER: Immanuel Williams, Ph.D.
Assistant Professor of Statistics

ABSTRACT

Is the Hot Hand Real? Evidence from a Permutation and Hierarchical-Based Analysis

Cameron An

The hot-hand phenomenon, often described as the tendency for individuals to experience prolonged streaks of success that exceed what would be expected under random performance, has been widely studied across many disciplines, particularly basketball. Early studies attempted to evaluate this effect through various techniques, often concluding that the hot-hand phenomenon was largely a myth. However, recent studies have begun to revisit previous analyses using improved statistical techniques, with some claiming evidence of a discernible hot-hand effect. This study examines the presence of the hot-hand effect in the modern NBA by testing whether observed shooting patterns deviate from those simulated under independence. A permutation-based analysis, which preserves observed shot outcomes while randomizing order, reveals that the observed shot sequences generally do not exhibit elevated streak behavior, providing limited evidence of a hot-hand effect beyond what would be expected by chance. To further account for player and season-level heterogeneity, a logistic hierarchical model was constructed to control for in-game variables that contribute to shot success. The estimated effects of being in a "hot" state are small and reinforce the findings of the permutation test after adjusting for additional covariates.

Keywords: Hot hand, Streak shooting, NBA, Permutation test, Logistic hierarchical model.

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Chapter 1

INTRODUCTION

The concept of the hot hand in basketball has been fiercely debated among audiences spanning many decades. Fans and players often describe the phenomenon as reaching an elevated state of play following a streak of recent success. One general notion of hot-hand shooting posits that players tend to make a higher percentage of shots relative to their baseline shooting ability following a recent streak of made shots. Proponents of the hot hand argue that the physical and psychological benefits of witnessing a streak of success place players “in rhythm”, a state of increased confidence and ability that may increase the chances of making a shot. Former Golden State Warriors player Purvis Short described the experience of being “hot” as follows: “You’re in a world all your own. It’s hard to describe. But the basket seems to be so wide. No matter what you do, you know the ball is going in” (Tversky & Gilovich, 1989). However, opponents argue that these experiences may reflect cognitive biases, suggesting that apparent streaks may arise purely from random variation. These perspectives highlight the intuitive appeal of the hot-hand phenomenon, raising the question of whether perceived streaks in performance reflect underlying changes in ability under unique circumstances or simply arise from chance.

Chapter 2

BACKGROUND

2.1 Previous Literature

Statistical research has questioned whether the hot hand truly exists, despite widespread belief from fans, players, and the general public. In a seminal study, Gilovich, Vallone, and Tversky (1985), examined streak-shooting tendencies to determine if observed streaks exceeded those expected under randomness. The sample data consisted of field goal attempts from 48 home games of 76ers players and their opponents during the 1980-81 season. The first section of their analysis used conditional proportions to examine players' shooting percentages conditional on the outcome of previous shots. For each of nine players on the 76ers, Gilovich, Vallone, and Tversky computed the overall shooting proportion as well as conditional shooting proportions based on whether the previous one, two, or three shots were successful. They also estimated the serial correlation between the outcomes of successive shots to determine if there existed any dependence structure within a player's shot sequence. A "success", or "hit", was defined as a made shot. The authors discovered that only one out of the nine sampled players possessed a higher success rate on shots following a make compared to their overall success rate. Furthermore, eight out of nine players exhibited a negative serial correlation value, none of which being significantly different from zero.

The findings in this paper contributed to the characterization of the hot hand as a myth, which researchers attribute to a cognitive fallacy whereby individuals misinterpret random streaks as performance trends (Gilovich, Vallone, & Tversky, 1985). A potential explanation for such a phenomenon involves memory bias, the idea that observers are likely to overestimate the dependence of sequential shots due to long sequences of successes (or failures) being more memorable than alternating successes and failures (Gilovich et al., 1985). Other studies have

151 since expanded upon the analysis of Gilovich, Vallone, and Tversky, yielding similar results
152 of a lack of hot-hand effect (Koehler & Conley, 2003; Lantis & Nesson, 2021; McNair,
153 Margolin, Law, & Ritov, 2020).

154 However, recent research has revisited the validity of the methods used in Gilovich, Vallone,
155 and Tversky (1985). In particular, Miller and Sanjurjo (2018) identified a flaw in the
156 conditional proportion analysis used in the 1985 paper. The flaw stems from a systematic
157 bias in the analysis of conditional shooting percentages within finite shot sequences, which
158 ultimately resulted in an underestimation of the true hot hand effect. Miller & Sanjurjo
159 (2018) formally demonstrated the downward bias in conditional proportion estimators by
160 analyzing all possible permutations of short finite sequences of independent binary outcomes.
161 In their example using three consecutive coin flips, they examine the proportion of heads
162 following a previous head under the assumption of independence. A calculation of the
163 conditional proportion estimator revealed that the expected proportion of heads conditioned
164 on a previous head is equal to $\frac{5}{12}$, notably lower than the suspected 0.5, illustrating the
165 systematic downward bias introduced by conditioning on finite sequences (Miller & Sanjurjo,
166 2018).

167 Miller and Sanjurjo applied the coin-flipping framework to demonstrate bias in the method-
168 ology of Gilovich, Vallone, and Tversky (1985), whereby the conditional proportions sys-
169 tematically underestimated the true hot hand effect of the data. Coin flips that resulted in
170 heads were treated as made shots, while coin flips that resulted in tails were treated as missed
171 shots. Applying the same logic to a basketball context, Miller and Sanjurjo reasoned that
172 finite samples of shot data exhibited the same bias observed in the coin problem. More
173 specifically, the expected proportion of made shots conditioned on a previous make was
174 not equal to the player's overall shooting percentage, but instead was biased downwards
175 due to the structure of the sampling process. To fix this issue, Miller and Sanjurjo utilized a
176 permutation test, described in detail in Section 4.1. After accounting for the downward bias in

177 conditional proportions with the permutation test, Miller and Sanjurjo discovered a reversal
178 of Gilovich's findings, indicating a discernible hot-hand effect for the given data.

179 **2.2 Contextual Variation in Hot-Hand Evidence**

180 While much research has been conducted on hot-hand trends, the settings of each analysis
181 provide little overlap and, therefore, are applied in different contexts. In addition to Gilovich's
182 analysis of 76ers players during the 1980-81 season, an analysis of conditional proportions
183 under a controlled shooting experiment with the Cornell intercollegiate men's and women's
184 basketball teams revealed that the outcomes of shot attempts were largely independent of the
185 outcome of the previous attempt (Gilovich et al., 1985). The controlled experiment required
186 players to shoot under highly regulated conditions, along with monetary incentivization
187 determined by the accuracy of their shots. These results were ultimately subject to the same
188 downward bias introduced by Miller and Sanjurjo.

189 Other analyses have yielded different results utilizing alternative methods and settings. An
190 analysis of serial dependence on 110,513 free throws in the NBA revealed a discernible,
191 albeit small, hot-hand effect in free throws (Mews & Ötting, 2023). This analysis accounted
192 for unevenly spaced time points between free throws by using a state-space model that
193 assesses a player's form using the Ornstein-Uhlenbeck process. Data from the NBA Three-
194 Point Contest also provide evidence of a hot-hand effect using the permutation testing
195 procedure presented by Miller and Sanjurjo (Ross, 2017). The aforementioned analyses
196 attempt to isolate the hot-hand effect by considering shots under controlled conditions to
197 prevent additional covariates from influencing the results. While this approach yields strong
198 results, they may not generalize to real game situations.

199 Additional research has offered unique perspectives on hot-hand trends by incorporating
200 context into a player's shots. An analysis of over 83,000 real game shots from the 2012-
201 13 NBA season revealed that players shooting over their baseline shooting percentage in

recent shots face different circumstances compared to when they are in their baseline state (Bocskosky, Ezekowitz, & Stein, 2014). These circumstances include longer shot distances, facing tighter defense, increased likelihood of taking their team's next shot, and overall attempting more difficult shots. The same study also determined that players in a "hot" state are more likely to make their next shot after controlling for shot difficulty.

Basketball philosophy has also evolved substantially since Gilovich's analysis in 1985, when offensive play was largely characterized by interior scoring. As player skills have evolved, teams have increasingly emphasized perimeter play and distance shooting (Wang & Zheng, 2022). In addition, factors such as pace of play may further influence shooting patterns. Together, these changes in offensive strategy and shot selection suggest that conclusions drawn from earlier periods may not fully generalize to the context of modern basketball, potentially altering the interpretation of the hot-hand effect.

Ultimately, the presence of the hot-hand phenomenon has largely remained unsolved due to differences in empirical research and statistical examinations of simulated data, which often rely on distinct assumptions and methodologies (Bar-Eli, Avugos, & Raab, 2006). Additionally, no clear definition of the hot hand exists, with studies varying in how they define and measure streak-shooting behavior. These discrepancies have contributed to ongoing debate regarding the validity of the effect.

2.3 Motivation

Many hot-hand analyses have been conducted in controlled settings or within specific game contexts, such as the NBA Three-Point Contest (Lantis & Nesson, 2024; Ross, 2017) or free throw shooting (Arkes, 2010; Chang, 2019; Lantis & Nesson, 2021). Many of these foundational studies were also conducted in a different era of the NBA than in Gilovich, Vallone, and Tversky (1985), when the style of play differed substantially from today's game. These environments offer valuable insights in their respective contexts, but may differ in

227 applicability to modern in-game playstyles.

228 The NBA has undergone notable changes in offensive philosophy from past eras, with a
229 greater emphasis on perimeter shooting, increased pace of play, and the development of
230 more versatile offensive skill sets. As a result, modern gameplay reflects a substantially
231 different environment than previous eras. Therefore, reexamining the hot hand using modern,
232 in-game data provides a new perspective on the presence and magnitude of streak shooting.
233 Specifically, this study examines the hot-hand effect across players and seasons in the modern
234 NBA, with the goal of evaluating streak-shooting tendencies over the past 10 seasons.

3.1 Modern NBA Trends

All data and figures presented in this section are derived from publicly available NBA statistics obtained from Basketball Reference (“NBA League Averages - Per Game,” n.d.).

The most notable example of the evolution of offensive strategy is Stephen Curry’s 2015-16 unanimous MVP season, in which he attempted a then-record 886 three-pointers. Figure 3.1 below displays an increasing trend in the percentage of total field goal attempts (FGA) that were three-point attempts (3PA) after the 2015-16 regular season. The season variable is encoded as the ending year of each NBA season (e.g., 2016 corresponds to the 2015-16 season).

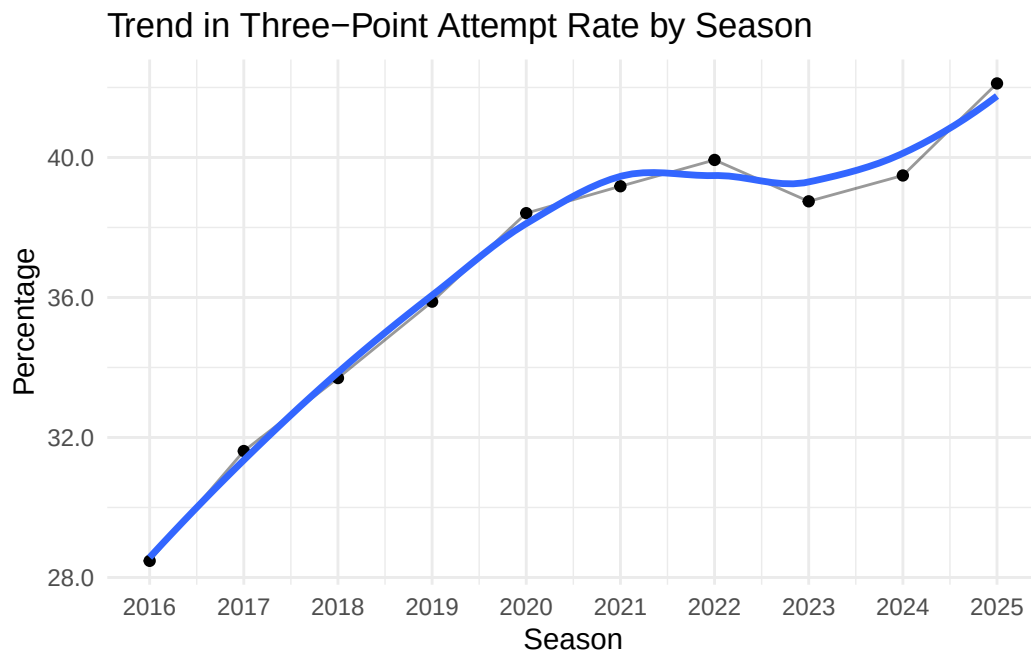


Figure 3.1: Increasing trend in three-point share by season.

246 The increase in three-point shooting raises the question of whether the hot-hand effect has
247 evolved over time. Improvements in offensive skill could amplify streakiness, whereas the
248 lower success rate of three-point attempts relative to two-point attempts may attenuate such
249 effects.

250 Since the 2015-16 regular season, teams have also emphasized a faster pace of play to get
251 more offensive possessions in a game. Pace, defined as the number of possessions a team
252 uses in a standard 48-minute game, heavily influences the number of shots taken within
253 a game and could have an impact on streak shooting. Figure 3.2 below depicts a marked
254 increase in the league-average pace from the 2015-16 to 2019-20 regular season, followed
255 by a gradual decline and stabilization in recent seasons.

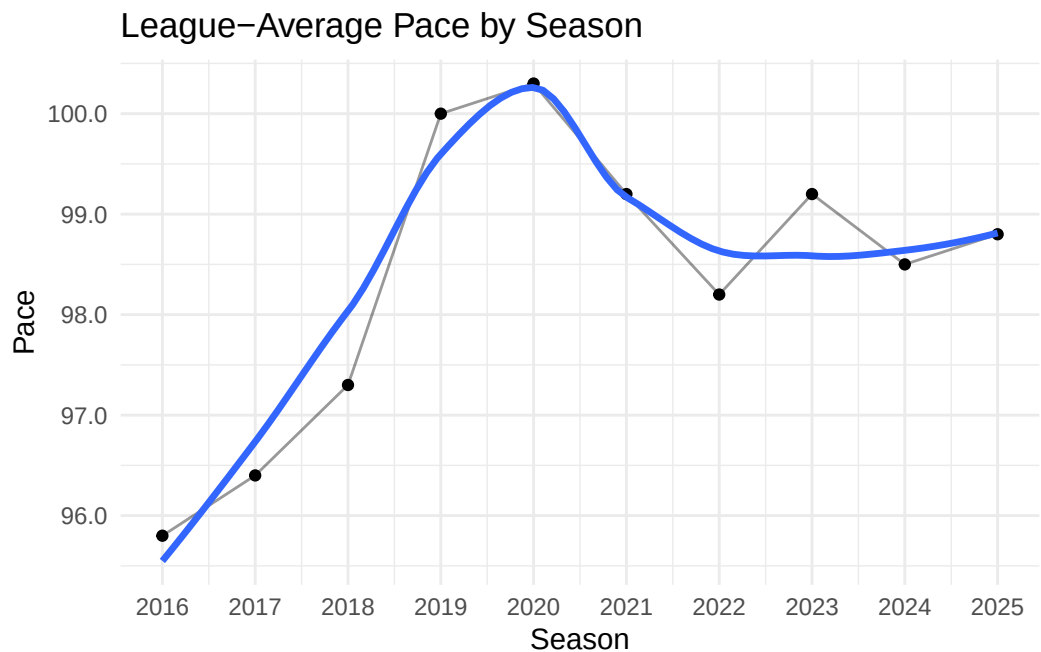


Figure 3.2: League-average pace varies by season, experiencing rapid change after 2016.

256 The fluctuations in pace since 2016 may suggest changes in streak-shooting tendencies
257 over time, as players attempt more shots per game and therefore have more opportunities to
258 develop scoring streaks.

259 Figure 3.3 displays the league-average field goal attempts (FGA) per game at the team level

260 increasing at a commensurate rate to the pace of play from approximately the 2015-16 to
 261 the 2019-20 regular season. Field goal attempts are defined as “any live-ball throw, shot, or
 262 tap—including layups and dunks—directed at a player’s own basket to score, excluding free
 263 throws” (*RULE NO. 4*, n.d.).

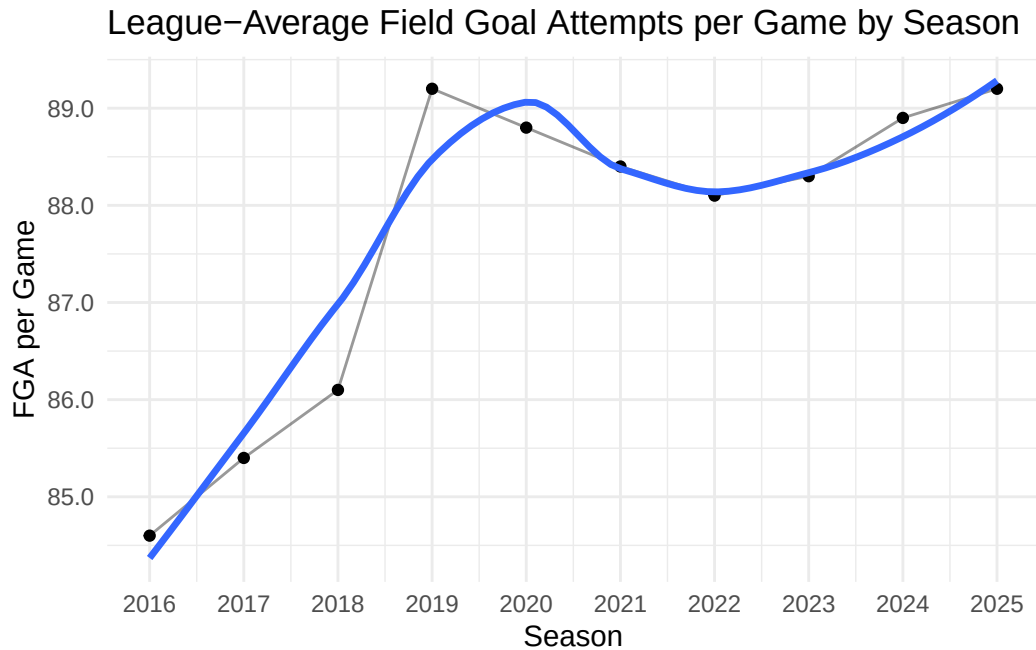


Figure 3.3: League-average FGA per game at the team level increasing after 2016.

264 The increase in field goal attempts per game coincides with the increase of pace, suggesting a
 265 fundamental alteration in offensive philosophy that may influence streak-shooting tendencies.

266 3.2 Data Collection

267 3.2.1 Sample Selection

268 Data on the top 50 players with the most field goal attempts per season were collected for
 269 each of 10 seasons from 2015-16 to 2024-25 (not including playoffs), yielding a total of 500
 270 player-seasons. A player-season is defined as a single season of performance for an individual
 271 player (e.g. 2015-16 Stephen Curry, 2019-20 LeBron James). Restricting the sample to
 272 the top 50 shooters per season focuses the analysis on high-volume players, who account

for a large share of their team’s shot attempts and are often among the most influential contributors. This approach reduces the overall sample size of the dataset while preserving a high number of shot attempts, allowing for higher statistical power in detecting a hot-hand effect. Importantly, some players were top 50 shooters in FGA in multiple seasons from 2015-16 to 2024-25. These players are represented multiple times in the analysis with unique shot sequences for each respective season in which they were a top 50 shooter. The 2015-16 season was chosen as the starting point due to its significance in changing league-wide trends in shot selection. It is of interest to analyze how the hot hand has progressed since that season, as more teams attempt to replicate the success of prolific offensive teams from that time frame.

All data were collected using the R package *hoopR* (*hoopR Package - RDocumentation*, n.d.), which provides access to NBA play-by-play statistics through SportsDataVerse API. Shot sequence data were obtained for the 500 player-seasons in consideration, with each sequence recording the sequential outcomes of all field goal attempts by a player within a player-season. For example, a shot sequence of ‘0011’, where 0’s represent missed shots and 1’s represent made shots, corresponds to a sequence in which a player missed the first two shots and made the last two shots. A missed shot was also defined as a “failure”, whereas a made shot was defined as a “success”. In total, data from 35,428 games across the 500 player-seasons were collected.

3.2.2 Exploratory Data Analysis

The distribution of the total number of field goal attempts per season for the top 50 shooters is displayed in Figure 3.4 below. Notably, the 2019-20 and 2020-21 seasons exhibit lower overall field goal attempts relative to other seasons, which is likely attributable to COVID-19-related restrictions that reduced the number of games played. Prior to those seasons, the distributions of field goal attempts are bimodal, with the larger peaks occurring at approximately 1,000 field goal attempts and the smaller peaks under 1,500 field goal attempts.

299 After the 2019-20 and 2020-21 seasons, the distributions become more unimodal, with peaks
 300 shifting to greater than 1,000 field goal attempts. This pattern could suggest a delayed impact
 301 of the 2015-16 season, when long-distance shooting and faster pace became the primary
 302 emphasis for most teams. In general, we see an increase in field goal attempts post-COVID,
 303 which could alter hot-hand trends.

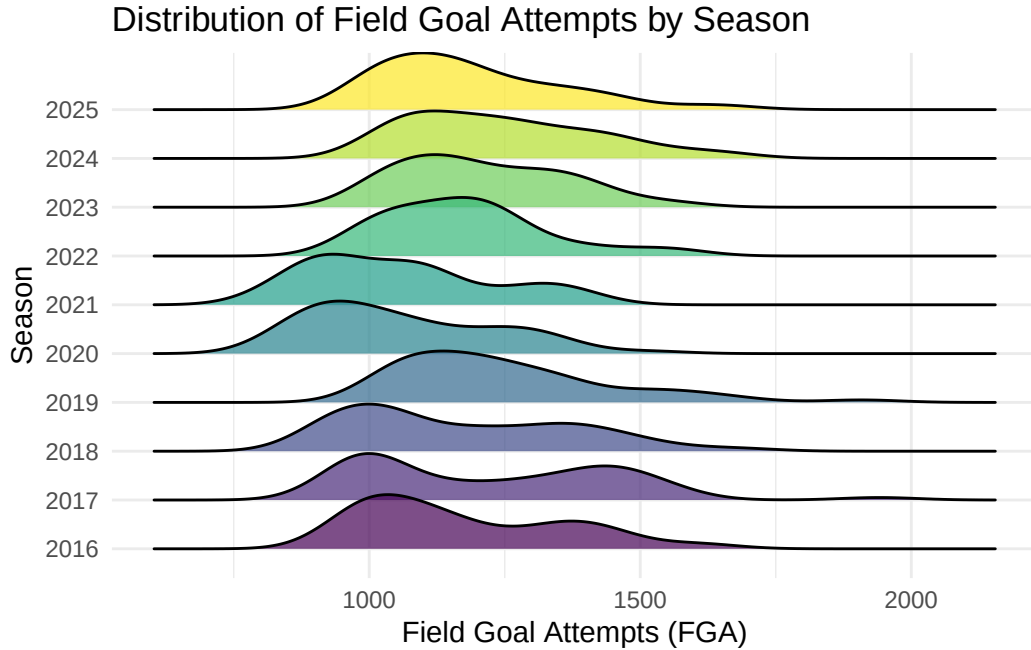


Figure 3.4: Distributions of FGA per game vary by season; shortened seasons in 2020 and 2021.

304 For each shot sequence, multiple variables were collected or constructed to help analyze the
 305 hot-hand effect. An *opportunity* is defined as any shot preceded by at least k consecutive
 306 made shots, and is used as an indicator of whether a player is on a hot streak. In this analysis,
 307 $k = 3$ is used to remain consistent with prior hot-hand literature.

308 For example, for $k = 3$, consider the sequence:

309 $0, 1, 1, 1, \underline{1}, \underline{0}, 1, 1, 1, \underline{1}$

310 Shots that occur immediately after three consecutive made shots are classified as opportunity
 311 shots and are underlined in the example above. These opportunity shots are used to evaluate

312 player performance while in a “hot” state.

313 An initial analysis of the data reveals that the mean number of opportunities per game for
314 qualified player-seasons fluctuates across seasons, albeit at small increments, as shown in
315 Table 3.1 below. This trend does not necessarily imply any increase in hot-hand shooting
316 ability, as the increase could stem from a variety of factors. As discussed earlier, league-wide
317 average field goal attempts per game have also increased over time, which could lead to
318 a higher number of observed opportunities simply due to a greater volume of shots being
319 taken.

Table 3.1: Number of Opportunities per Game per Player by Season

Season	Mean Opportunities
2016	1.12
2017	1.34
2018	1.29
2019	1.36
2020	1.31
2021	1.47
2022	1.36
2023	1.53
2024	1.58
2025	1.42

320 To adjust for the increase in the number of field goal attempts taken per game, the percentage
321 of total field goal attempts that were opportunity shots was calculated and plotted in Figure
322 3.5. The percentage of opportunity shots shows an overall upward trend across the period, but
323 with noticeable year-to-year fluctuations. This increasing trend suggests a higher prevalence
324 of streak occurrences over the past 10 seasons, which could be attributed to changes in

325 offensive strategies or players becoming more capable of entering hot streaks. Importantly,
 326 this trend only describes a slight increase in the percentage of opportunity shots as opposed
 327 to an actual hot-hand effect.

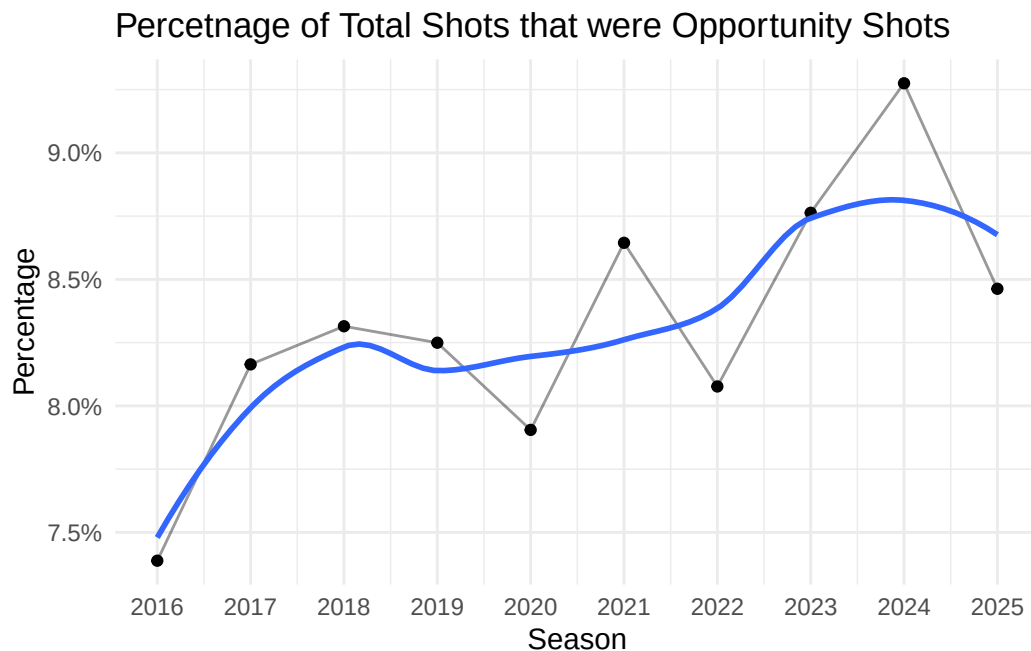


Figure 3.5: Proportion of Opportunity Shots showing general upward trend for qualified player-seasons.

328 Opportunity field goal percentage, defined as the field goal percentage on opportunity shots,
 329 for qualified player-seasons has also generally increased over time, as shown in Figure 3.6.
 330 The graph starts at its lowest point during the 2015-16 season and increases steadily until the
 331 2018-19 season, where it initially peaks. The 2019-20 season displays a steep decline in hot
 332 hand field goal percentage, but eventually recovers and continues to rise until it peaks during
 333 the 2023-24 season. The wave-like pattern of the smoothed line suggests variability over
 334 time rather than a strictly monotonic increase. An upward trend provides some preliminary
 335 evidence of an increase in hot-hand shooting ability over time, but needs to be formally
 336 tested for statistical significance as it may be attributable to random variation.

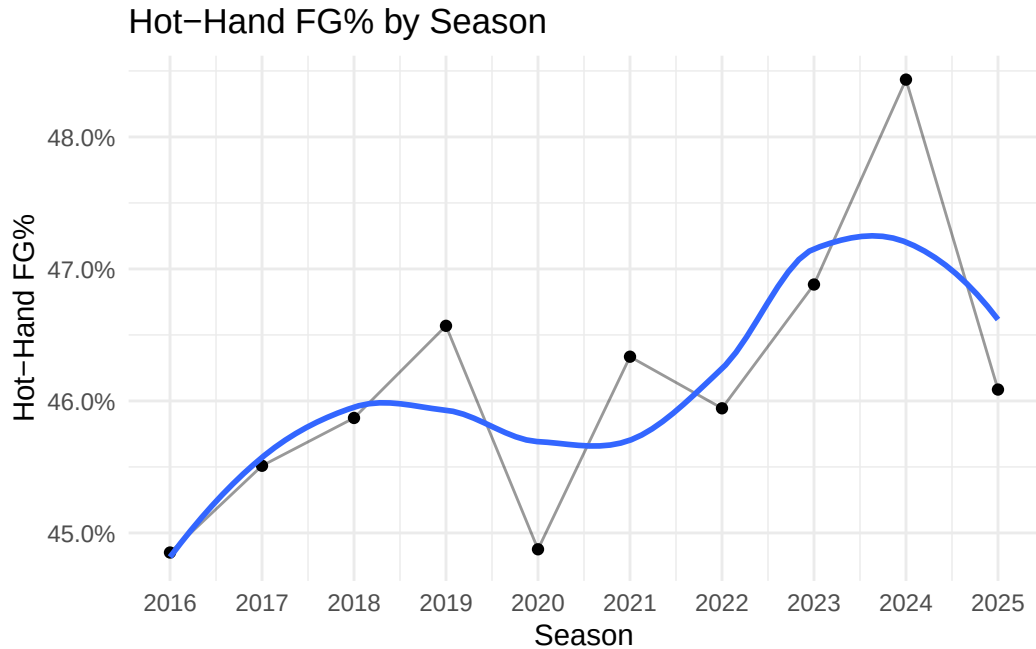


Figure 3.6: Hot-hand FG% showing general upward trend over time for qualified player-seasons.

Although the trends discussed above may indicate a higher degree of streakiness, a meaningful evaluation requires a comparison to baseline shooting performance. Figure 3.7 below addresses this by plotting the difference between opportunity FG% and baseline FG% for all qualified player-seasons from 2015-16 to 2024-25. For each season, the baseline FG% is computed by summing the total number of made field goals from qualified players during that season divided by the total number of field goal attempts during that season from qualified players. This provides a natural benchmark with which the opportunity FG% can be evaluated.

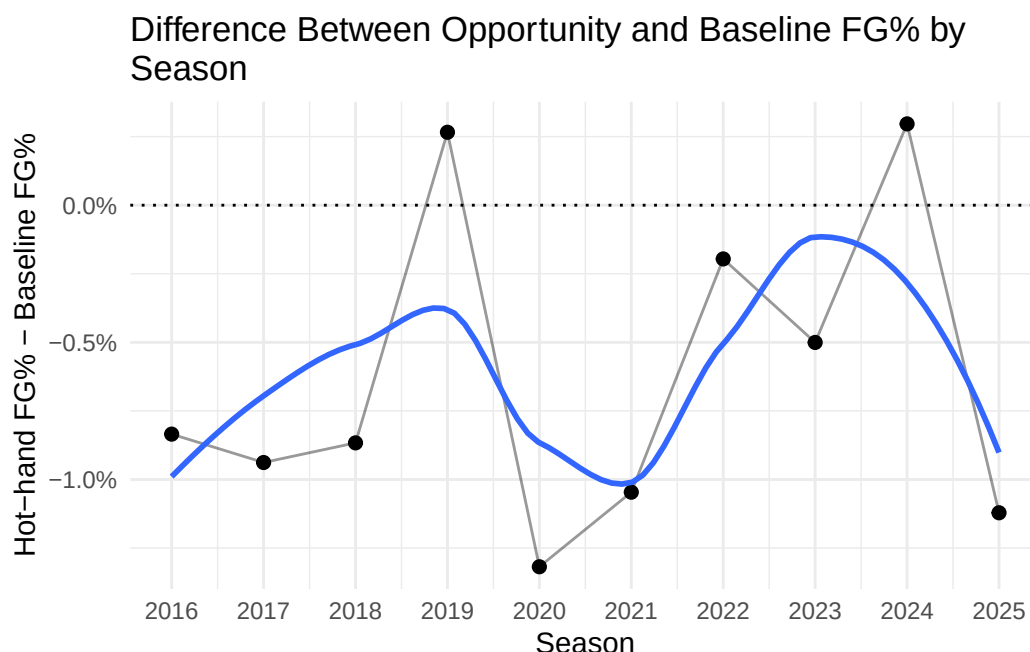


Figure 3.7: Difference between Opportunity and Baseline FG%, with horizontal marker at 0%.

345 The overall difference seems to fluctuate around zero, with most seasons possessing a slight
 346 negative difference, indicating that the success rate on opportunity shots is less than the
 347 success rate for baseline attempts on average. While there are isolated seasons, such as
 348 2018-19 and 2023-24, in which hot-hand FG% is greater than the baseline FG%, these
 349 deviations are neither persistent nor large in magnitude. These results do not provide any
 350 preliminary evidence that a systematic hot hand effect exists on the aggregate level. It is
 351 important to note that this analysis remains subject to the selection bias identified by Miller
 352 and Sanjurjo, as the hot-hand FG% is computed by conditioning on prior shot outcomes.
 353 Therefore, more rigorous and detailed analysis methods that account for this bias are needed
 354 before any conclusions can be made regarding the hot-hand effect.

355 The distance of each shot attempt was collected using hoopR's coordinate system, with
 356 each shot containing an x and y-coordinate relative to the basket. Shot distances were then
 357 organized into one of three categories: close range (0-8 feet), mid-range (8-22 feet), and

three-pointer (22+ feet). These category labels are approximate, as the distance of a three-point attempt varies depending on court location. For example, corner three-point attempts are taken from approximately 22 feet, whereas three-point attempts taken from the top of the arc must be at least 23.75 feet. The chosen cutoff values reflect the empirical distribution of shot distances, which tend to cluster in specific regions. Figure 3.8 below displays the distribution of shot distances for all games within the 500 qualified player-seasons, with vertical lines marked at cutoff values of 8 and 22 feet.

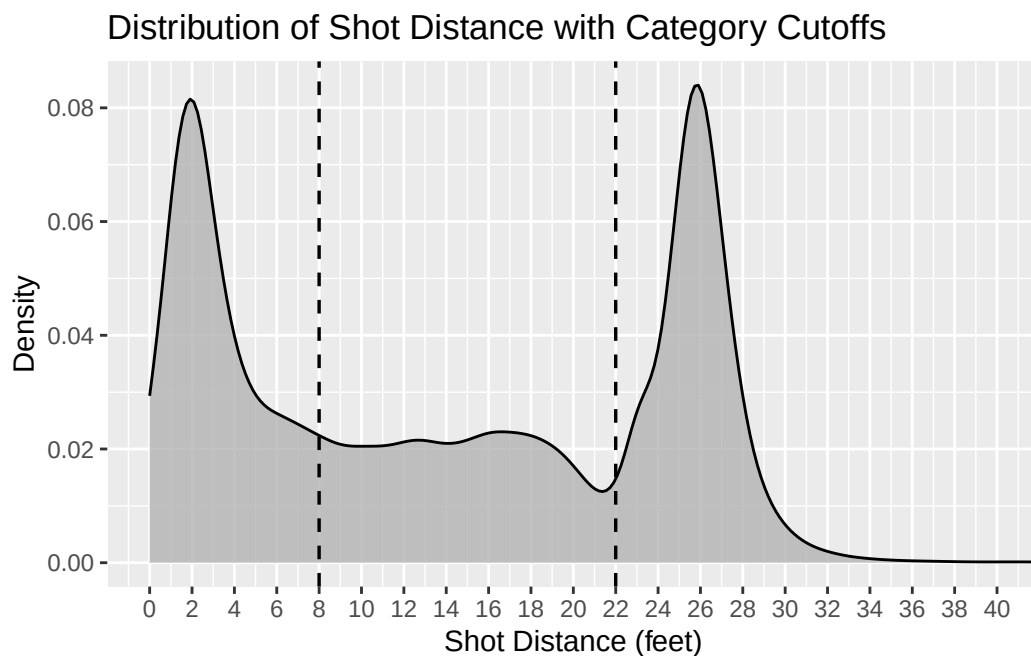


Figure 3.8: Distribution of Shot Distances with cutoff values at 8 and 22 feet.

The density plot exhibits clear clustering of shots near the basket (approximately 0-8 feet) and beyond the three-point line (approximately 22-30 feet), with relatively fewer attempts in the mid-range. This varying structure suggests that teams tend to compartmentalize shot types into various categories rather than a continuous spectrum. Therefore, modeling shot distance as a categorical predictor with three distinct levels provides a more practical and interpretable representation of shot selection than treating it as a purely continuous quantitative variable.

3.3 Streak Statistics

Ross (2017) outlines various *streak statistics* that analyze finite sequences of binary outcomes. A streak statistic is a measure designed to quantify the “streakiness” of a given shot sequence. The streak statistic used in my analysis is defined below:

Difference in proportion of S (opportunities - non-opportunities): The difference between the proportion of success on opportunity shots and the proportion of success on non-opportunity shots. In the previous example, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, the value of the streak statistic was $\frac{2}{3} - \frac{6}{7} \approx -0.1905$ (Ross, 2017).

This statistic was chosen because it provides a direct comparison of shooting performance in a “hot” state against shooting performance outside of that state. Additionally, it has an intuitive interpretation under the assumption of independence, as it is often expected to be equal to zero. However, as discussed earlier, this expectation is biased downward in finite samples, which will be revisited in the subsequent analysis.

Chapter 4

PERMUTATION TESTING

4.1 Permutation Testing Procedure

The permutation testing procedure utilizes a streak statistic to assess the likelihood of observing a shot sequence at least as extreme as that computed from the observed shot sequence. We define the null and alternative hypotheses:

H_0 : Shots are independent.

H_a : Shots are dependent.

The steps of the procedure are outlined below:

1. Calculate the streak statistic for the observed shot sequence.
2. Randomly permute the outcomes of the observed shot sequence while preserving the total number of successes and failures. This produces a sequence consistent with the null hypothesis of independent shot attempts.
3. Calculate the streak statistic for the permuted shot sequence.
4. Repeat steps 2-3 many times (e.g. 10,000 iterations) to approximate the sampling distribution of the streak statistic assuming independence of shot attempts.
5. Calculate the proportion of permuted streak statistics that are at least as extreme as the observed statistic in the direction consistent with the hot-hand effect to obtain the p -value.

This procedure addresses the bias identified in Gilovich, Vallone, & Tversky (1985) by directly constructing an appropriate null distribution from the observed shot sequence rather

than relying on conventional assumptions under independence. The random permutation of observed outcomes satisfies the null assumption of independence, allowing us to compare the observed shot sequence to many sequences simulated under the null hypothesis.

4.2 Season-Level and Within-Game Shot Sequences

The permutation testing procedure was applied to season-long shot sequences as well as within-game shot sequences to assess the robustness of results across different levels of aggregation. Each approach provides distinct advantages that address limitations inherent in the other.

4.2.1 Season-long Data

Each observation in this sample consists of an individual player's sequential shot data for a particular season. That is, for each of the 500 player-seasons, a binary sequence of makes and misses is constructed using all shots taken over the course of the season. Shots are carried over across games, meaning they accumulate over the course of a season without resetting after each game. As a result, streaks can extend across game boundaries. For instance, if a player makes the final two shots of one game and the first shot of the next game, the subsequent shot would be classified as an opportunity. Treating each game separately, as examined in the within-game analysis, effectively censors shot sequences at the end of each game by truncating potential streaks. This could result in weaker evidence of a hot-hand effect by reducing the number of observed opportunity situations and inflating p-values. By allowing sequences to carry over across games, this analysis avoids inflating p-values and maximizes the statistical power under the given context.

The permutation test is applied to each of these season-long sequences, allowing for the evaluation of streak behavior over a larger sample of shots per player. Using season-long shot data allows for greater statistical power due to the large number of opportunities within a given sequence. With more qualifying observations, estimates of conditional shooting

431 probabilities become more stable and less sensitive to random variation, increasing the
432 ability to detect meaningful deviations from the null hypothesis. It also offers a unique
433 perspective on hot-hand shooting, with results applying to prolonged stretches of time rather
434 than only individual games. Significant results for season-long data could indicate a potential
435 carry-over effect, with players experiencing increased levels of play across multiple games
436 given that the player finished the previous game on a streak.

437 Appendix A displays the permutation test results for the 30 player-seasons exhibiting the
438 strongest evidence of a hot-hand effect.

439 At an $\alpha = .05$ individual significance level, only 10 out of 500 player-seasons exhibit
440 a statistically discernible hot-hand effect. However, when conducting 500 independent
441 hypothesis tests, we would expect approximately $0.05 * 500 = 25$ false rejections purely
442 by chance, even if no true hot-hand effect exists. Thus, the observed number of significant
443 results is lower than what we would expect under the null hypothesis of independence. Figure
444 4.1 below displays the distribution of p-values for the 500 player-seasons. The distribution
445 is skewed left, with many observations possessing p-values greater than 0.05.

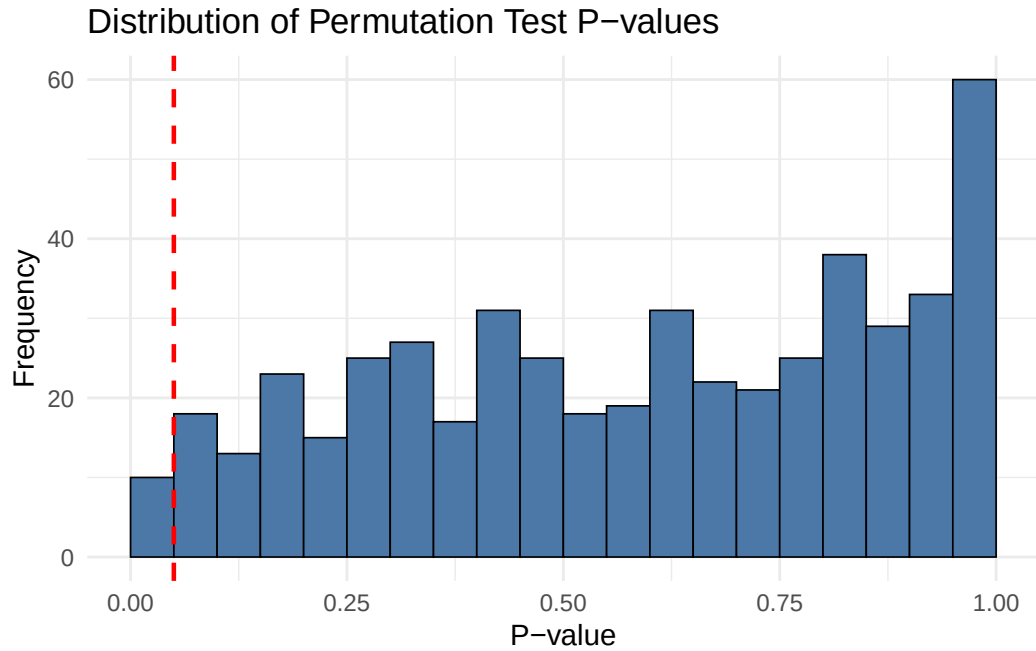


Figure 4.1: Distribution of Permutation Test P-values displaying left skew.

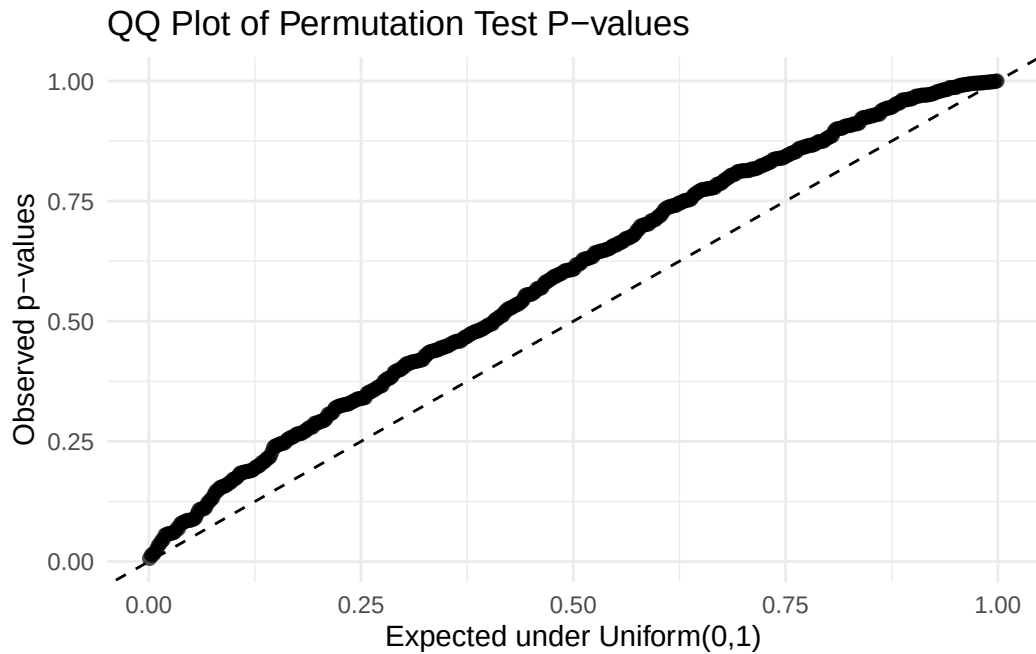


Figure 4.2: Distribution of Permutation Test P-values compared to uniform dsitribution.

446 The QQ Plot (Figure 4.2) above compares the observed p-values from the permutation tests
 447 to the uniform distribution. Under the null assumption of independence, we expect the

distribution of p -values to follow a uniform distribution, displayed by the diagonal line. The hot-hand effect suggests that the distribution of observed p -values should fall below the diagonal line since a larger proportion of player-seasons would be expected to show stronger evidence than under independence, resulting in lower quantiles. However, the observed p -values align above the diagonal line, which not only suggests that there is not only little evidence of a hot-hand effect, but also that players potentially perform slightly worse during opportunities than other shots. These results are aggregated at the player-season level, based on season-long shot sequences rather than game-level data. As a result, the analysis captures within-season patterns but may mask potential short-term dynamics.

To account for the false discovery rate of hot-hand effects, the Benjamini-Hochberg procedure was applied to the existing p -values. After this adjustment, none of the player-seasons remain statistically significant, providing no evidence of a hot-hand effect after correcting for multiple comparisons.

4.2.2 Game-level Data

Season-long shot sequences were then deconstructed further into game-level sequences. In this sample, each observation corresponds to a binary sequence of makes and misses for a game within a player-season. Thus, each player-season contributes multiple sequences, one for each game in which the player recorded field goal attempts. The permutation test is applied to each game-level shot sequence separately, capturing within-game streak behavior and allowing for a more granular analysis of hot-hand shooting patterns. Results for game-level data are often considered more practical for hot-hand analysis than season-long data since most people attribute the hot-hand effect to individual games rather than prolonged periods of time. However, using game-level data decreases the statistical power of the permutation test considerably, as shot sequences typically contain very few opportunities, if any.

473 In total, 35,428 shot sequences were considered for the game-level analysis. The distribution
 474 of total FGA per game for qualified players is displayed below in Table 4.1 and Figure 4.3,
 475 with a mean of 16.5 FGA per game and a standard deviation of 5.19 FGA per game.

Table 4.1: Total Shots per Game per Player

Minimum	Q1	Median	Mean	SD	Q3	Maximum
1	13	16	16.5	5.19	20	50

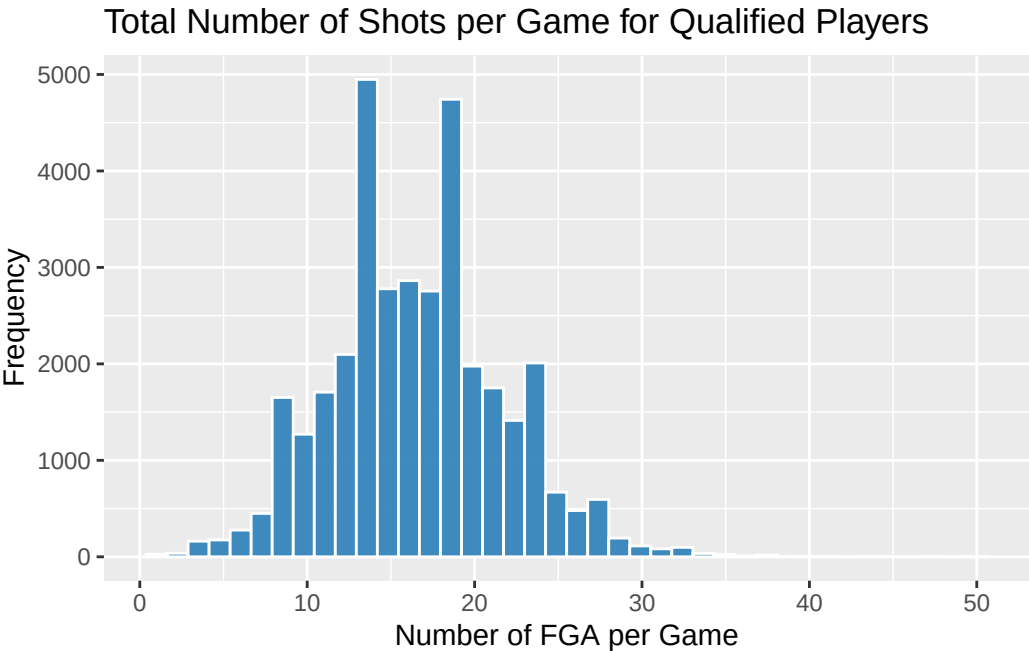


Figure 4.3: Total shots per game for qualified player-seasons.

476 The limited sample size of each shot vector restrains the number of opportunity shots within
 477 a sequence, thereby reducing the statistical power of the permutation test. In addition, 14,563
 478 games (approximately 41.1%) in the sample contain no opportunity shots which further
 479 reduces the effective sample size to 20,865 games. Games with zero opportunity shots were
 480 retained in the dataset but classified as providing no information for the permutation test.

481 **4.3 Quantifying the Bias**

482 Prior work has shown that the magnitude of small-sample bias decreases as sample size
483 increases (Miller & Sanjurjo, 2018). Thus, the magnitude of bias for the season-long data is
484 minimal and has little effect on the observed streak statistic. In contrast, the limited sample
485 size of shot sequences in the within-game analysis warrant a closer examination of this bias.
486 Because the magnitude of the bias is not immediately clear, a simulation-based approach
487 was used to quantify its impact under realistic conditions for the within-game analysis.

488 **4.4 MCMC Simulation**

489 A Monte Carlo simulation was used to approximate the expected value of the streak statistic
490 *Difference in proportion of S (opportunities – non-opportunities)* under different sequence
491 lengths and success probabilities. A streak length of three was used for the simulation. The
492 estimation procedure is summarized in Algorithm 1.

Algorithm 1 MCMC for Expected Value of Streak Statistic

1: **Input:** Sequence lengths $n \in \{5, 6, \dots, 50\}$, streak length $k = 3$, probability $p \in \{0.40, 0.45, 0.50, 0.55, 0.60\}$, number of simulations M

2: **Output:** Expected value $\mathbb{E}[\hat{p}_{\text{opportunity}} - \hat{p}_{\text{others}}]$ for each (n, p)

3:

4: **for** each n **do**

5: **for** each p **do**

6: Initialize list $T = \emptyset$

7: **for** $j = 1$ to M **do**

8: Generate $X_1, X_2, \dots, X_n \sim \text{i.i.d. Bernoulli}(p)$

9: $S \leftarrow \emptyset$ $\triangleright$ indices of streak-following shots

10: **for** $i = k + 1$ to n **do**

11: **if** $X_{i-k} = X_{i-k+1} = \dots = X_{i-1} = 1$ **then**

12: $S \leftarrow S \cup \{i\}$

13: **end if**

14: **end for**

15: **if** $S = \emptyset$ **or** $S^c = \emptyset$ **then**

16: **continue**

17: **end if**

18: $\hat{p}_{\text{opportunity}} \leftarrow \frac{1}{|S|} \sum_{i \in S} X_i$

19: $\hat{p}_{\text{others}} \leftarrow \frac{1}{|S^c|} \sum_{i \notin S} X_i$

20: $T_j \leftarrow \hat{p}_{\text{opportunity}} - \hat{p}_{\text{others}}$

21: $T \leftarrow T \cup \{T_j\}$

22: **end for**

23: ExpectedValue(n, p) $\leftarrow \frac{1}{|T|} \sum T$

24: **end for**

25: **end for**

26: **Return:** {ExpectedValue(n, p)}

Figure 4.4 below displays the expected value of the streak statistic using Monte Carlo simulation. The parameter p varies from 0.40 to 0.60, in increments of 0.05, reflecting plausible shooting proportions observed in game settings, although more extreme values exist due to small sample sizes. The overall sample proportion of made shots across all 35,428 qualified games was approximately 0.467. For each (n, p) pair, the simulation was conducted using 10,000 trials and the resulting expected values are presented in the figure below.

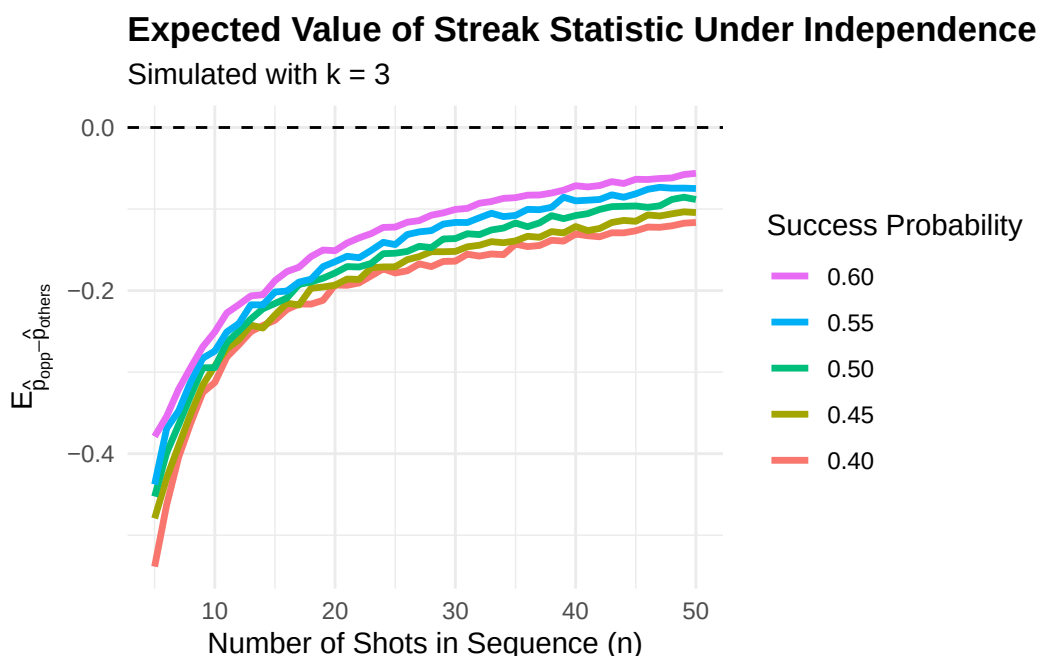


Figure 4.4: Simulated bias of streak statistic under different conditions.

The curves exhibit a consistent vertical ordering with higher success probabilities corresponding to a smaller magnitude of bias. Differences in bias across success probabilities appear relatively consistent, indicating that the bias generally scales proportionally with p . Additionally, each line appears to be monotonically increasing towards 0, meaning that the expected bias decreases as n increases for a given success probability p .

These results can be compared to the null statistics of each observed shot sequence generated by the permutation test. However, it is important to note that shot sequences typically vary

with p and n , which would alter the magnitude of bias for that particular sequence.

4.5 Bias-Adjusted Streak Statistic

The permutation test was run for all shot sequences in the sample. Sequences with zero opportunities were assigned a p -value of NA . Appendix B contains a sample of 30 games, stratified based on the number of total shots, and the collected statistics through the permutation tests. To visualize the effect of different sample sizes on the null distribution, three games of different sample sizes were collected. The three selected games contained 13, 16, and 20 shots to represent Q1, the median, and Q3, respectively. Each game had a streak statistic of -0.5 for consistency. The observed and null statistics are shown in Table 4.2 below.

Table 4.2: Bias-Adjusted Streak Statistic for Different Sample Sizes

Player	Total Shots	Total Makes	Opportunity %	Difference in Proportions	Null Mean	Bias-Adjusted Difference
Kyle Kuzma	20	9	0	-0.5	-0.235	-0.265
Tobias Harris	16	7	0	-0.5	-0.242	-0.258
Jordan Clarkson	13	6	0	-0.5	-0.292	-0.208

The *Difference in Proportions* column represents the streak statistic, defined as the difference between the proportion of makes on opportunity shots and the proportion of makes on all other shots. However, this statistic is subject to the bias discovered by Miller & Sanjurjo (2018). The *Bias-Adjusted Difference* column corrects this statistic by subtracting the null mean obtained from the observed streak statistic. This adjustment accounts for bias in the

original streak statistic, yielding a bias-adjusted streak statistic.

As expected, the null means generally seem to decrease as the number of shots increase, although some trend deviation remains due to differences in field goal percentages across games (Figure __, Appendix). The average null mean across all games with at least one opportunity was approximately -0.229, indicating a substantial bias in the streak statistic. This estimated bias agrees with the results obtained from the Monte Carlo simulation, as our overall success probability was approximately 0.467 and the average number of shots taken per game was approximately 16.5. The null standard deviations are relatively large compared to the season-long analysis, reflecting the high variability inherent in short shot sequences. Because many games contain a limited number of opportunities, the field goal percentage on those opportunities can vary substantially, leading to highly volatile streak statistics. For example, in games with only one opportunity, the opportunity field goal percentage can take on only two values (0 or 1). As a result, the streak statistic is highly discrete, and the corresponding permutation distribution is characterized by a small set of possible values that can differ markedly from one another, contributing to considerable variability in the null distribution. Consequently, games with few opportunities will also have lower statistical power, making it more difficult to detect meaningful hot-hand trends.

4.6 Game-Level Results

Overall, only 562 out of 20,865 games (approximately 2.69%) that included at least one opportunity produced a statistically significant streak statistic at the $\alpha = 0.05$ significance level. This proportion is lower than what would be expected under the assumption of independence, providing little evidence in support of a within-game hot-hand effect. Furthermore, the mean of the *Adjusted Difference in Proportions* column across all games with at least one opportunity was approximately -0.0136, indicating that, on average, players performed slightly worse on opportunity shots than would be expected under independence.

Figure 4.5 provides the distributions of the bias-adjusted streak statistic by season. The distributions appear largely consistent across seasons, exhibiting similar bimodal shapes and comparable peak locations. In particular, each season shows two prominent peaks centered at approximately -0.25 and 0.25. The black points denote the seasonal means of the null-adjusted streak statistic, which are generally centered slightly below zero. This aligns with the previous finding that players tend to perform marginally worse on opportunity shots relative to other shots after adjusting for the bias. One exception is the 2018-19 season in which the mean appears slightly above zero, suggesting a small, though likely negligible, positive hot-hand effect. The similarity between the distributions suggests little evidence of systematic seasonal differences in hot-hand effects.

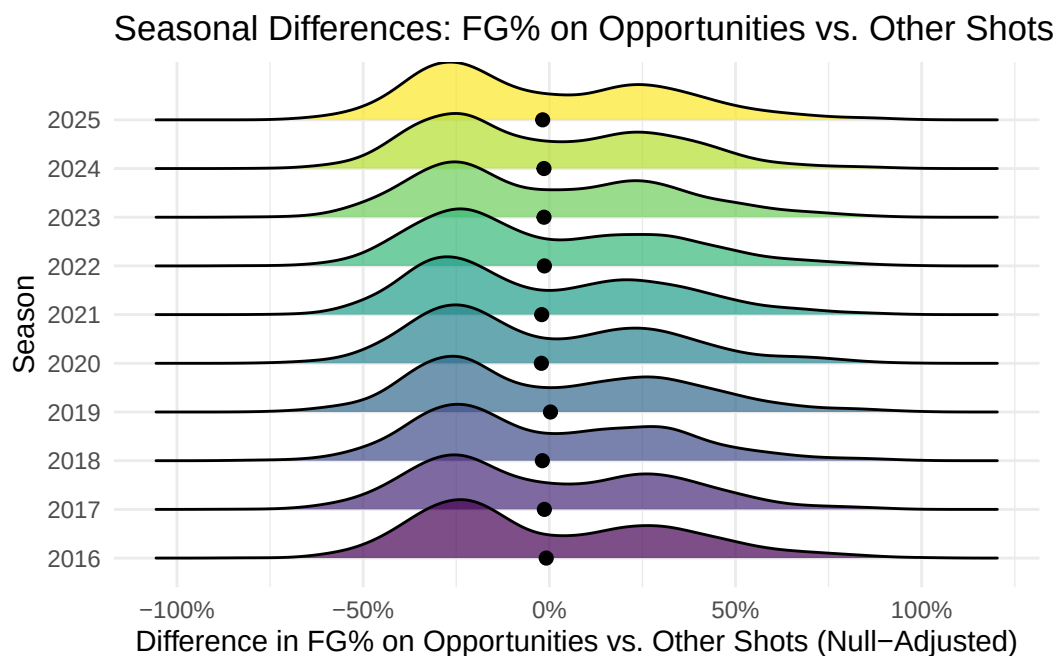


Figure 4.5: Bias-Adjusted Streak Statistic Distributions by Season.

The observed bimodality can be explained by the structure of the streak statistic that results from small shot sequences. Because many games contain few opportunities, the opportunity field goal percentage is often either very low (e.g., 0) or moderate (e.g., around 0.5), while the field goal percentage on other shots is typically near 0.5, reflecting both the larger number

561 of non-opportunity (“other”) shot attempts in each game and the fact that league-wide
 562 field goal percentages tend to be close to this value. Consequently, the unadjusted streak
 563 statistic frequently takes values near -0.5 or 0. Figure 4.6 displays this pattern by using the
 564 three previously sampled games which contained different numbers of total shots (n) and
 565 successful shots (m). The null distributions reveal points clustering at approximately -0.5
 566 and 0, confirming our previous intuition.

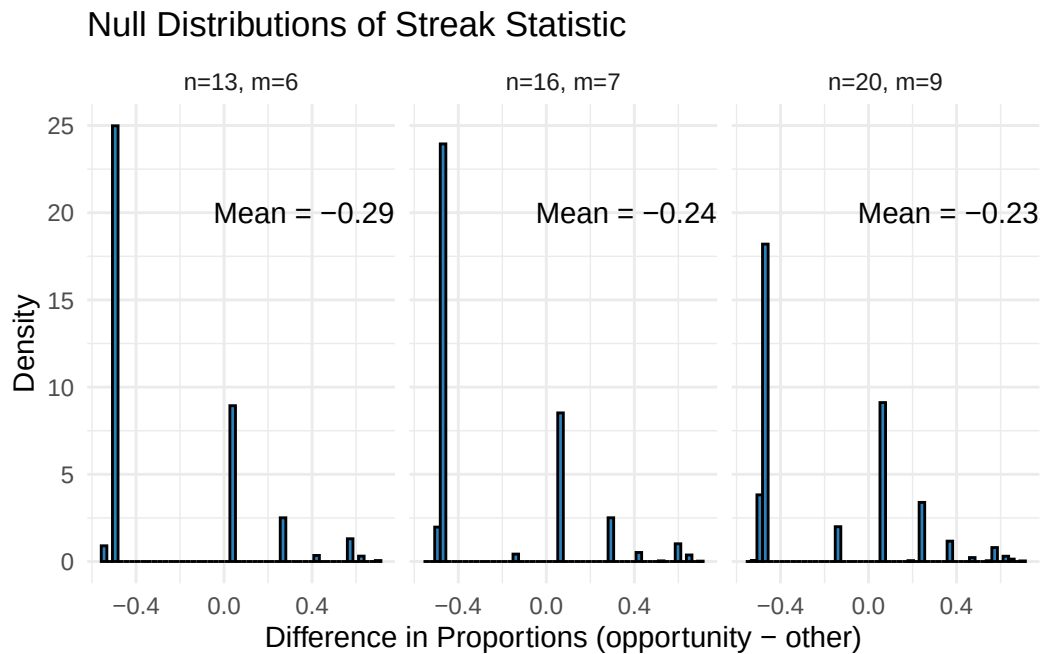


Figure 4.6: Null Distributions of Streak Statistic under Different Scenarios.

567 After adjusting for the null mean (approximately -0.23), these values shift to around -0.25
 568 and 0.25, producing the observed peaks. This pattern reflects the high variability of the
 569 statistic rather than systematic differences across seasons, providing little evidence of a
 570 meaningful season-specific hot-hand effect.

571 To examine the results further, the games were categorized by the number of opportunities
 572 in their respective sequence in Table 4.3. The table shows an overall negative relationship
 573 between the number of opportunities within a game and the number of games associated
 574 with the number of opportunities. This is expected because qualifying for opportunity shots

is increasingly difficult as the number of opportunities increases. The *Proportion of Games* with $Adj_Diff > 0$ column represents the proportion of games in which the bias-adjusted streak statistic was greater than 0, indicating a positive directional effect in favor of the hot hand.

Table 4.3: Proportion of Games with Bias-Adjusted Difference in Proportions Greater than 0.

Number of Opportunities	Number of Games	Proportion with Positive Adjusted Difference
0	14563	NA
1	8184	0.043
2	5401	0.676
3	3341	0.644
4	1912	0.763
5	984	0.774
6	545	0.813
7	282	0.791
8	122	0.779
9	59	0.881
10	16	0.688
11	12	0.750
12	3	0.667
13	3	0.333
14	0	NA
15	1	1.000

The proportion of games with adjusted difference in proportions is undefined for games with exactly 0 or 14 opportunities. Games with zero opportunities do not yield a streak

581 statistic, and zero games in the sample contain exactly 14 opportunities. Among games with
582 at least one opportunity, those with only one opportunity comprise approximately (39.22%),
583 representing the largest share across all opportunity counts. However, only about 4.3% of
584 these games exhibit a positive null-adjusted difference in proportions, indicating minimal
585 evidence of a hot-hand effect for single-opportunity games.

586 This result is consistent with the structural constraints of the shot sequence and qualifications
587 of an opportunity shot. For a game to contain exactly one opportunity, the player must miss
588 the opportunity shot. Otherwise, a made opportunity shot would generate an additional
589 opportunity on the subsequent attempt. Consequently, the only scenario in which a positive
590 adjusted streak statistic can occur in a one-opportunity game is when the lone opportunity
591 arises at the end of the sequence, such as when the player makes the final four shots of
592 the game with no prior opportunities. In this case, the final shot constitutes a successful
593 opportunity while precluding the possibility of additional opportunities. Because this scenario
594 occurs infrequently, the proportion of games with a positive adjusted streak statistic is
595 correspondingly low, as reflected by the 4.3% figure.

596 In games with more than one opportunity, the proportion of games exhibiting a positive
597 adjusted streak statistic is substantially larger than in single-opportunity games, with all
598 but the 13-opportunity category exceeding 50%. This pattern can be explained by the
599 underlying structure of the data, as multiple opportunities within a game typically arise
600 from prior successful opportunity shots, leading to higher opportunity field goal percentages
601 and, consequently, larger adjusted streak statistics. For instance, consider a shot sequence
602 with five opportunity shots. Due to the limited number of shots that a player can take
603 within a game, it is very likely that the opportunity shots clump together, forming a string of
604 successful opportunity shots. The example shot sequences below illustrates this concept,
605 with each containing five opportunity shots.

606

Sequence 1:

607

0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0

608

Sequence 2:

609

0, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0

610

Sequence 3:

611

0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0

612 The difference in proportions between opportunity and non-opportunity shots for each se-
 613 quence s ven by:

614 [Sequence 1: $\frac{4}{5} - \frac{5}{10} = 0.3$] [Sequence 2: $\frac{3}{5} - \frac{6}{10} = 0$] [Sequence 3: $\frac{1}{5} - \frac{12}{20} = -0.4$]

615 Sequences 1 and 2 contain opportunity shots that are more tightly clustered, necessitating
 616 a sequence of successful opportunity shots. This results in higher opportunity field goal
 617 percentages, which ultimately causes the streak statistic to increase. In contrast, sequence
 618 3 contains opportunity shots that are more dispersed throughout the sequence, indicating
 619 more unsuccessful opportunity shots. This structure decreases the field goal percentage on
 620 opportunity shots, resulting in a lower streak statistic.

621 Importantly, sequences similar to 1 and 2 are more likely to occur in real game data, as the
 622 number of shots taken during the duration of a game are limited and multiple opportunities
 623 typically arise from consecutive made shots. This structural feature likely contributes to the
 624 increased number of games with positive bias-adjusted streak statistics for games with a
 625 large number of opportunity shots.

626

Chapter 5

627

LOGISTIC HIERARCHICAL MODEL

628 While the permutations tests for both season-long and within-game sequences provide little
629 evidence of a hot-hand effect, these analyses do not account for contextual factors that may
630 influence shooting performance. Incorporating additional variables into our analysis allows
631 for a reexamination of the hot-hand effect while accounting for specific in-game conditions.

632 **5.1 Shot Distance Analysis**

633 Shot distances were analyzed to assess whether players exhibit different shooting distances
634 on opportunity shots compared to non-opportunity shots. Summary statistics for quantitative
635 shot distance by opportunity status are presented in Table 5.1, where “all” includes both
636 opportunity and other shots.

Table 5.1: Average Shot Distance by Shot Type.

Shot Type	Average Distance (Ft.)
All	14.28
Opportunity	16.03
Other (non-opportunity)	14.13

637 The table reveals that opportunity shots are taken from a greater average distance than
638 other shots, suggesting increased difficulty in these situations. Figure 5.1 displays the
639 distribution of shot distances by shot type, with both curves exhibiting a similar bimodal
640 shape corresponding to close-range and three-point attempts. However, while the peaks of
641 the “other” shot distribution are of roughly equal height, the opportunity shot distribution

642 features a more pronounced peak at longer distances compared to its peak at shorter distances.
 643 This shift indicates that opportunity shots are more likely to be taken from longer distances
 644 than other shots, reinforcing the pattern observed in Table 5.1.

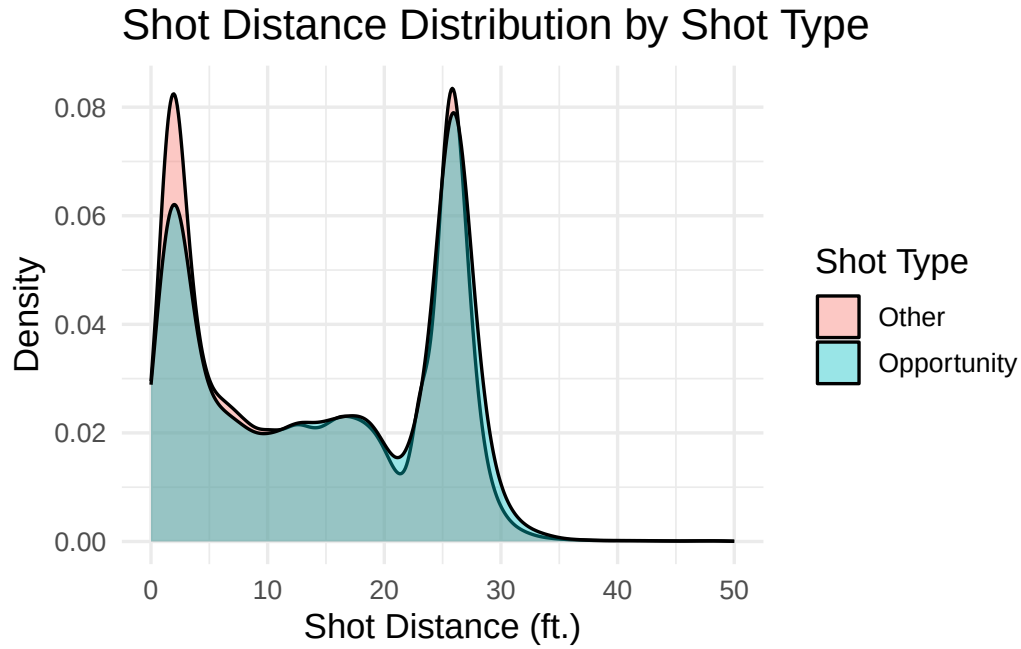


Figure 5.1: Shot Distances by Shot Type.

645 To further solidify the differences between opportunity and non-opportunity (other) shots,
 646 the average shot distance for each shot type was computed within each player-season to
 647 account for player-season variation. For each player-season, the difference between mean
 648 opportunity shot distance and mean non-opportunity shot distance was calculated. The mean
 649 of these differences was approximately 2.14 feet, indicating that, on average, opportunity
 650 shots were taken from a greater distance than other shots within the same player-season.

651 A paired t -test was conducted to assess whether the population mean difference in shot
 652 distance was greater than zero. The results provide strong evidence that opportunity shots
 653 are taken from a greater distance than non-opportunity shots, $t(499) = 30.94, p < 0.001$.
 654 The estimated mean difference was 2.14 feet, with a one-sided 95% confidence interval
 655 indicating that the true mean difference is at least 2.03 feet.

Figure 5.2 presents a scatterplot of average shot distance on opportunity shots versus average shot distance on non-opportunity shots for each player-season, grouped by position. Positions were categorized into guard, forward, and center based on classifications from Basketball Reference (“NBA League Averages - Per Game,” n.d.). The diagonal reference line represents equality in shot distance between the two shot types. The majority of points lie above this line, indicating that, for most player-seasons, opportunity shots are taken from greater distances than non-opportunity shots.

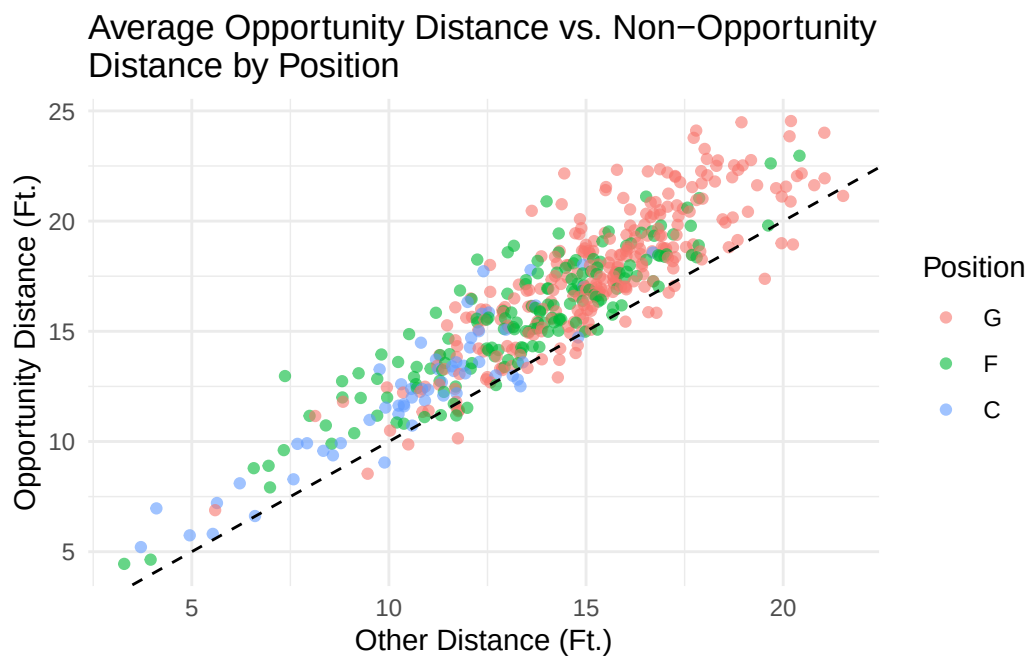


Figure 5.2: Average Shot Distances by Shot Type for Qualified Player-Seasons.

Shot distance was also analyzed on a categorical scale to account for the large proportion of shots taken from grouped distances. The three levels of shot distance are close (0-8 feet), mid-range (8-22 feet), and three-pointer (22+ feet), as outlined in Section 3.2.2.

Table 5.2: Proportions of Distance Category by Shot Type.

Shot Type	Close	Mid-range	Three-Pointer
Opportunity	0.333	0.286	0.380

Shot Type	Close	Mid-range	Three-Pointer
Other	0.376	0.283	0.341

Table 5.2, along with Figure 5.3, reveals that a larger proportion of opportunity shots were three-pointers compared to other shots, while a smaller proportion were close shots. The proportion of mid-range shots remained relatively similar between shot types. A chi-square test of independence confirmed that the distribution of distance categories is dependent on shot type $\chi^2(2) = 420.55, p < .001$.

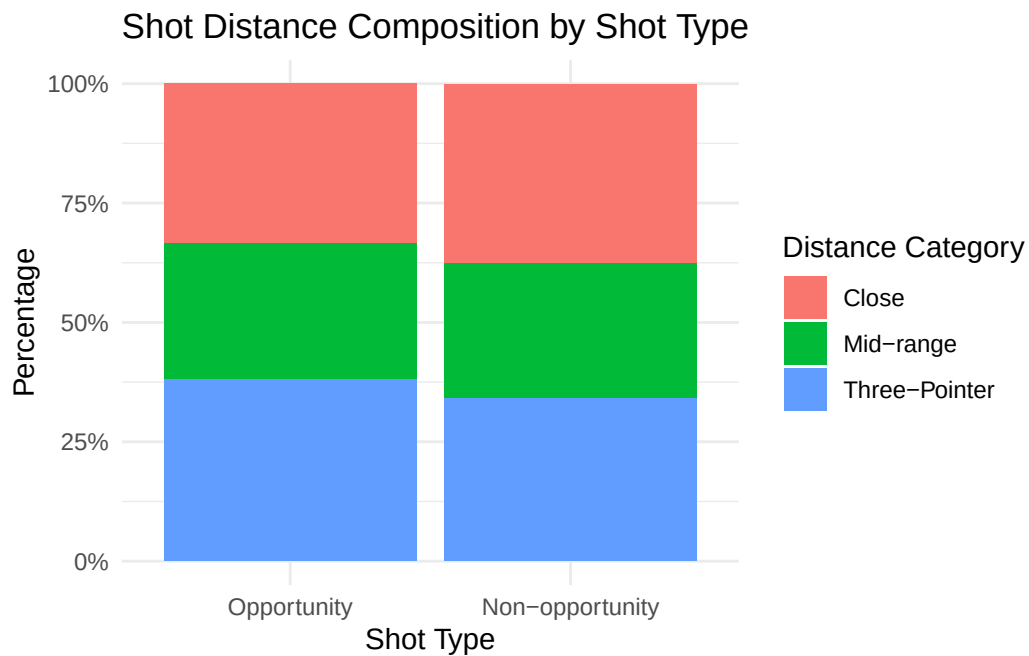


Figure 5.3: Distance Category by Shot Type.

This phenomenon likely arises from a combination of psychological and strategic factors. Players may describe the sensation of being “on fire”, referring to a mental and physical state of heightened focus and elevated performance. This can lead to increased confidence and manifest in more aggressive shot selection, including attempts from greater distances. The general public commonly refers to this state as heat-check behavior, as players attempt to

676 capitalize on their recent success. Other possible explanations involve strategic adjustments
 677 from both teams. Defenses may adjust by applying tighter coverage to a perceived “hot”
 678 player, forcing shots from further distances or worse positions. In turn, offenses may settle
 679 for longer-distance shots, as these are often quicker to attempt or more readily available
 680 under increased defensive pressure.

681 Due to this phenomenon, a new model that includes shot distance as a variable would greatly
 682 improve the results of the analysis by further isolating the hot-hand effect.

683 **5.2 Model Specification**

684 A logistic hierarchical model was constructed to evaluate the probability of a successful shot
 685 based on specific in-game variables and random effects. The variables and random effects
 686 are outlined in Table 5.3.

687 Let y_{ijk} denote the outcome of attempt i by player j in season k , where $y_{ijk} = 1$ if the
 688 attempt is made and 0 otherwise. We model

$$y_{ijk} \sim \text{Bernoulli}(p_{ijk})$$

$$\begin{aligned} \text{logit}(p_{ijk}) = & \beta_0 + \beta_1 \text{opportunity}_{ijk} + \beta_2 \text{midrange}_{ijk} + \beta_3 \text{threepoint}_{ijk} \\ & + \beta_4 (\text{opportunity}_{ijk} \times \text{midrange}_{ijk}) + \beta_5 (\text{opportunity}_{ijk} \times \text{threepoint}_{ijk}) \\ & + u_{0j} + v_{0k} + u_{1j} \text{opportunity}_{ijk} \end{aligned}$$

$$u_{0j} \sim \mathcal{N}(0, \sigma_{u0}^2)$$

$$u_{1j} \sim \mathcal{N}(0, \sigma_{u1}^2)$$

$$v_{0k} \sim \mathcal{N}(0, \sigma_{v0}^2)$$

Table 5.3: Hierarchical Model Variable Descriptions.

Variable	Model Term	Description	Effect Type
make	y_{ijk}	Indicator for whether a shot was made (1 = make, 0 = miss)	Response
intercept	β_0	Baseline log-odds for the reference category	Fixed
is_opportunity	$\beta_1 \text{ opportunity}_{ijk}$	Opportunity shot indicator (1 = yes, 0 = no)	Fixed
Midrange	$\beta_2 \text{ midrange}_{ijk}$	Midrange shot indicator (1 = yes, 0 = no)	Fixed
ThreePoint	$\beta_3 \text{ threepoint}_{ijk}$	Three point shot indicator (1 = yes, 0 = no)	Fixed
is_opportunity:Midrange	$\beta_4 (\text{opportunity}_{ijk} \times \text{midrange}_{ijk})$	Interaction between opportunity and midrange	Fixed
is_opportunity:ThreePoint	$\beta_5 (\text{opportunity}_{ijk} \times \text{threepoint}_{ijk})$	Interaction between opportunity and three point	Fixed
Player	u_{0j}	Player-specific intercept	Random intercept
is_opportunity by Player	$u_{1j} \text{ opportunity}_{ijk}$	Player-specific opportunity effect	Random slope

Variable	Model Term	Description	Effect Type
Season	v_{0k}	Season-specific intercept	Random inter-cept

Hot-hand theory posits that the probability of a successful shot increases on opportunity shots compared to non-opportunity shots, which many attribute to a positive coefficient for the opportunity variable. However, this coefficient is subject to the small-sample bias discovered by Miller and Sanjurjo, which must be accounted for. The inclusion of this variable allows us to assess how the probability of a successful shot varies with opportunity status while simultaneously accounting for other factors.

Player and season random effects were also included in the hierarchical model to account for unexplained heterogeneity in shooting performance and opportunity effects. In particular, the model includes random slopes for the opportunity effect across different players rather than assuming a uniform effect. Allowing different slopes for opportunity effects allows us to see which players are most impacted, or potentially not impacted, by hot-hand opportunities. Season random intercepts were also included to account for differences in baseline shooting ability across seasons.

5.3 Coefficient Bias

The opportunity coefficient is prone to the small-sample bias present in permutation testing. Therefore, the coefficient becomes inflated and overestimates the true effect of an opportunity shot on success probability. To account for this, a random sample of 100 games was selected from the full dataset. The following steps were taken to obtain a null distribution of opportunity coefficients from the sample:

- 708 1. Randomly permute the shot sequences within the 100 sampled player-games to simu-
709 late independent shots. All variables pertaining to each shot should be held constant,
710 with only the order of shots being permuted.
- 711 2. Fit the logistic hierarchical model for the reordered data to obtain a null opportunity
712 coefficient.
- 713 3. Repeat steps 1-2 many times (e.g., 10,000 trials) to obtain a null distribution of
714 opportunity coefficients.
- 715 4. Calculate the mean of the null distribution to obtain the opportunity coefficient ex-
716 pected under independence.

717 The procedure described above contained only 100 games due to computational limitations.
718 Therefore, the sample may not fully represent the full dataset, but should provide a general
719 estimate for the opportunity coefficient expected under independence.

720 **5.4 Results**

721 The null distribution of opportunity coefficients for the 100 randomly sampled games is
722 displayed in Figure 5.4. The mean of this distribution provides an estimate of the opportunity
723 coefficient expected under independence and will be used to determine if the observed
724 coefficient exceeds what would be expected under independence.

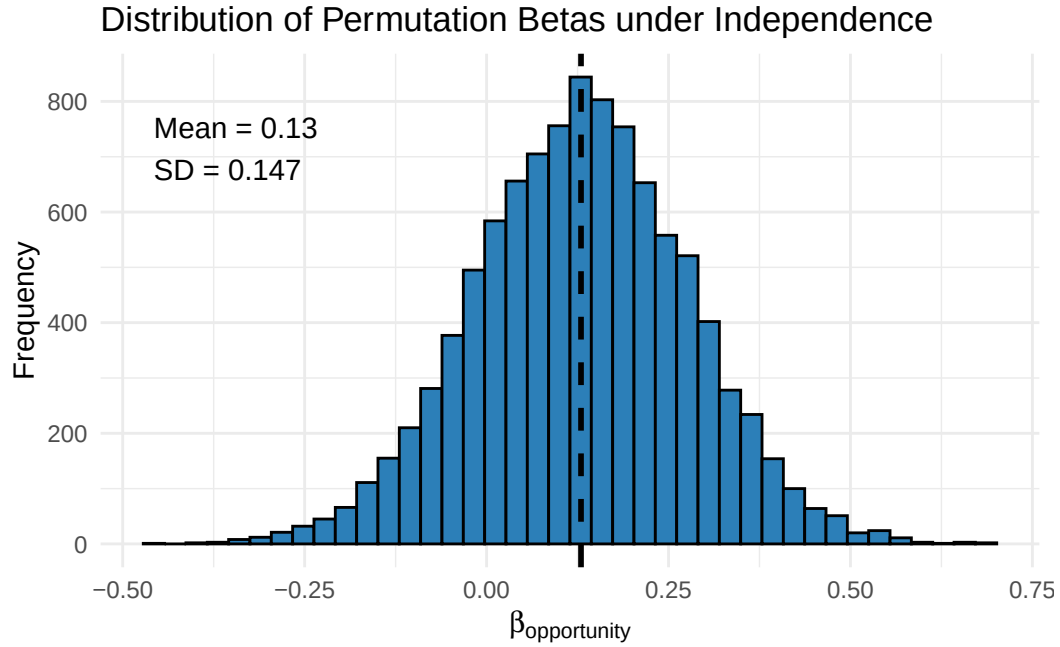


Figure 5.4: Histogram of Simulated Opportunity Coefficients under Independence.

725 The mean opportunity coefficient of the simulated trials under independence is approximately
726 0.13, suggesting the existence of a bias that overestimates the true hot-hand effect. We now
727 compare these results to the coefficients obtained from the observed hierarchical model.
728 The fixed coefficient estimates for the full data are listed in Table 5.4 and random effect
729 coefficients are listed in Table 5.5.

Table 5.4: Fixed Effect Coefficients.

Coefficient	Estimate	Std. Error	z value	p-value
Intercept	0.3207	0.0173	18.5423	0e+00
is_opportunity	0.0634	0.0168	3.7856	2e-04
Midrange	-0.6126	0.0070	-87.1207	0e+00
ThreePoint	-0.8927	0.0068	-130.3290	0e+00
is_opportunity:Midrange	-0.0826	0.0245	-3.3723	7e-04
is_opportunity:ThreePoint	-0.1224	0.0232	-5.2717	0e+00

Table 5.5: Random Effects (Standard Deviations)

Group	Effect	Std. Dev.
Player	Intercept	0.1056
Player	is_opportunity	0.0002
Season	Intercept	0.0449

730 The opportunity indicator variable yields a coefficient of 0.0634, which is notably lower than
 731 the expected opportunity coefficient under independence. This suggests that the estimated
 732 hot-hand effect in the hierarchical model is actually weaker than what would be expected if
 733 shots were independent. The statistically significant p -value for the opportunity coefficient
 734 in Table 5.4 should be interpreted with caution, as it is derived by comparing the estimated
 735 coefficient to 0. However, the simulation of null opportunity coefficients for the 100
 736 randomly sampled games yielded an estimated coefficient of 0.13, which represents the
 737 actual value that should be compared to our observed coefficient. Figure 5.5 displays the
 738 observed opportunity coefficient in relation to the null coefficients, along with the p -value
 739 describing the proportion of simulated trials with an opportunity effect as strong as the
 740 observed effect.

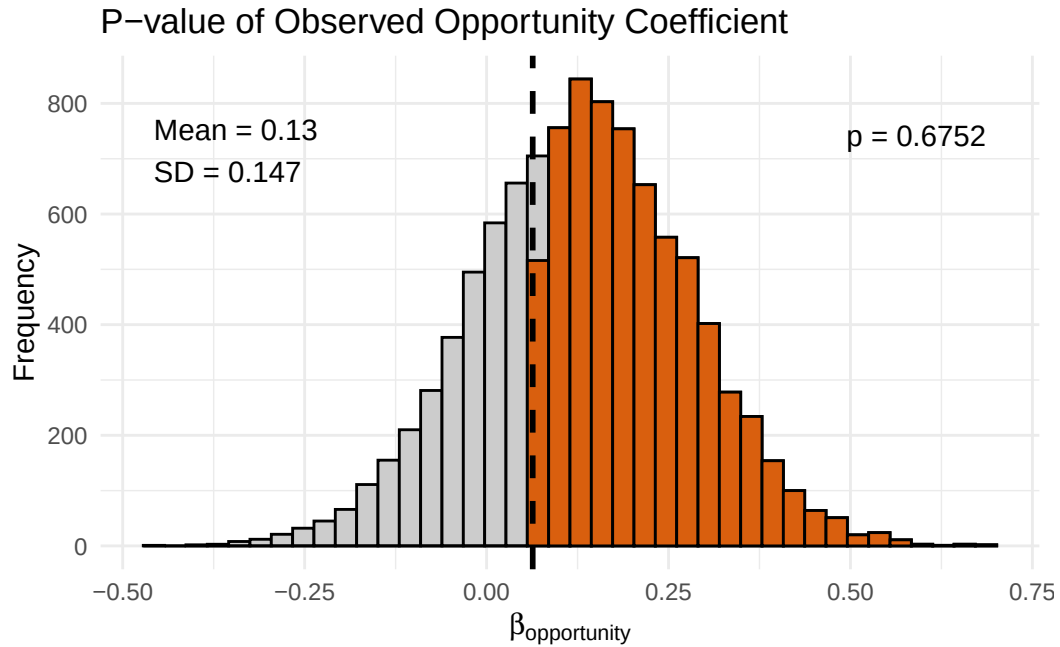


Figure 5.5: P-value of observed opportunity effect compared to null opportunity effects.

741 A large p -value of 0.6752 is obtained from the simulation, indicating weak evidence of
 742 an overall hot-hand effect over the past 10 seasons for high-usage players. The observed
 743 opportunity coefficient is consistent with what would be expected under the assumption of
 744 independence.

745 As expected, we also observe increasingly negative coefficients as shots are taken further
 746 from the basket. The mid-range and three-point indicators both show substantial decreases
 747 relative to close-range shots, reflecting the increase in shot difficulty with distance. The
 748 interaction between opportunity and distance is also negative for both mid-range and three-
 749 point shots, indicating that the impact of an opportunity shot decreases as shot distance
 750 increases.

751 The player-level random intercept yields a relatively large standard deviation, representing
 752 the substantial heterogeneity in baseline shooting ability across players. This is expected in
 753 basketball data, as skillsets vary drastically across players and contribute to different baseline
 754 shooting percentages. In contrast, the standard deviation of random slopes for the opportunity

755 indicator is extremely small, suggesting that the opportunity effect is relatively consistent
756 across players. The season-level random intercept is non-negligible, indicating that the
757 baseline shooting ability varies across seasons. This could reflect changes in offensive
758 strategy, shooting ability, or other contextual factors that could impact shooting efficiency
759 from year to year.

Chapter 6

CONCLUSION

6.1 Summary

The permutation tests for season-long shot sequences did not differ significantly from those expected under independence, offering little evidence of a hot-hand effect over the duration of a season despite possessing greater statistical power. Within-game shot sequences were also analyzed to examine the hot-hand effect on a more practical context, although with reduced statistical power due to smaller sample sizes. The within-game streak statistics did not differ significantly from those expected under independence, offering little evidence of a within-game hot-hand effect. Finally, a hierarchical model was constructed to evaluate the hot-hand effect while controlling for shot distance. Comparing the observed opportunity effect to null coefficients simulated under independence revealed that the opportunity effect does not differ significantly from those expected under independence, even after adjusting for shot distance.

6.2 Limitations

Importantly, the results obtained from the permutation test are specific to the sample of players considered, which primarily consist of high-usage players who attempt a large number of shots for their respective teams. As such, these findings do not generalize to low-usage players, whose shot selection and opportunities may differ substantially. Additionally, the permutation testing framework does not account for contextual factors that may influence the perception or existence of a streak.

Within-game shot sequences also present a challenge for detecting hot-hand patterns due

782 to limited sample sizes. The average number of shots per game for players in the sample
783 was approximately 16, resulting in relatively few opportunity shots and, consequently, low
784 statistical power. Considering season-long shot sequences increases the power of detecting
785 a hot-hand by increasing sample sizes and increasing the number of opportunities. However,
786 the results lose practical interpretability, as the hot-hand effect is generally understood to
787 occur within games rather than across extended periods.

788 Additional limitations arise from modeling assumptions and measurement choices. The hier-
789 archical model assumes independence between shots, which is partially addressed through
790 the inclusion of the opportunity variable. However, the opportunity variables does not fully
791 capture within-game dynamics that may induce dependence of shots. Furthermore, defining
792 an opportunity as three consecutive made shots serves only as a proxy for a “hot” state and
793 may not fully reflect the underlying hot-hand effect.

794 The bias-adjustment procedure in the hierarchical analysis used a sample of 100 games
795 to determine a null distribution of beta coefficients for the opportunity effect. While this
796 procedure simulates independent shot sequences, the resulting null distribution is specific to
797 the 100 randomly sampled games and may not fully capture the variability present in the
798 complete dataset. As a result, the estimated bias adjustment may be sensitive to the particular
799 sample of games selected, potentially limiting the generalizability of the findings.

800 **6.3 Future Work**

801 Additional analyses can be implemented to address the limitations presented above. Using
802 a larger sample of players would allow for greater generalization to the NBA as a whole
803 rather than only high-volume players. Incorporating additional variables, such as time
804 between shots, defensive pressure, and game situation would help to further isolate the
805 hot-hand effect and distinguish it from general shot streakiness. The permutation test and
806 hierarchical model in this analysis employed a streak length of 3, which limits the frequency

807 of opportunities. Shortening the streak length (e.g., $k = 1$) would increase the number of
808 opportunities within a shot sequence and, therefore, statistical power, but could weaken the
809 contextual interpretation of an opportunity as representing a genuinely “hot” state. Future
810 research could also extend this analysis to incorporate temporal dependence and a deeper
811 examination of what constitutes a “hot” state.

812 Increasing the number of sampled games would improve the bias adjustment for the hi-
813 erarchical model coefficient by providing a more stable estimate of the null opportunity
814 coefficients. Future analyses can look to construct hierarchical models on an individual
815 player level to account for the variation between players. The bias adjustment would then
816 pertain to specific players rather than the sample as a whole.

817 **6.4 Computational Details**

818 The results in this paper were obtained using R 4.5.3. R itself and all packages used are avail-
819 able from the Comprehensive R Archive Network (CRAN) at <https://CRAN.R-project.org/>.

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865

APPENDICES

866

Appendix A

Top 30 Player-Seasons with Largest Degree of Hot Hand (sorted by p-value of the permutation test)

Player	Season	Number of shots	Number of successes	Number of runs	Proportion of s on trials following streaks	Proportion of s on all other trials	Difference in Proportions	Null median	Null mean	Null SD	p-value	z-score
Kyle Lowry	2016	1195	502	578	0.531915	0.410336	0.121379	-0.003675	-0.005211	0.054369	0.0084	2.328351
Damian Lillard	2016	1457	601	698	0.508929	0.404461	0.104468	-0.001378	-0.002320	0.050061	0.0128	2.133171
Giannis Antetokounmpo	2021	1116	642	558	0.639269	0.559643	0.079626	-0.001361	-0.002483	0.038419	0.0128	2.137192
Shai Gilgeous-Alexander	2022	1051	477	511	0.544554	0.444211	0.100344	-0.003617	-0.004678	0.054159	0.0210	1.999160
Tobias Harris	2019	1253	611	600	0.559524	0.476498	0.083026	-0.003156	-0.003797	0.043946	0.0216	1.975685
Trae Young	2019	1276	534	624	0.505051	0.411215	0.093836	-0.002972	-0.003389	0.052923	0.0320	1.837083
Khris Middleton	2019	1161	513	555	0.521368	0.432950	0.088417	-0.002873	-0.002963	0.052827	0.0402	1.729805
Hassan Whiteside	2017	973	542	467	0.612022	0.544304	0.067718	-0.003358	-0.003527	0.042145	0.0434	1.690493
Keldon Johnson	2022	1011	471	482	0.539823	0.456570	0.083253	-0.002999	-0.003764	0.053323	0.0456	1.631883
Giannis Antetokounmpo	2023	1280	708	639	0.600917	0.543315	0.057603	-0.001051	-0.002348	0.037102	0.0490	1.615846
OG Anunoby	2025	915	425	458	0.535354	0.455882	0.079471	-0.003267	-0.004857	0.054778	0.0578	1.539446
Kemba Walker	2019	1693	733	815	0.490798	0.426797	0.064000	-0.002372	-0.003137	0.044507	0.0586	1.508464
D'Angelo Russell	2020	845	360	433	0.508475	0.419847	0.088627	-0.002480	-0.004867	0.064293	0.0610	1.454178
Nikola Vucevic	2016	1016	507	509	0.558140	0.490417	0.067722	-0.003312	-0.003741	0.047805	0.0616	1.494885
Kemba Walker	2018	1373	593	634	0.496000	0.425481	0.070519	-0.003605	-0.003682	0.049272	0.0628	1.506166
Karl-Anthony Towns	2017	1479	801	744	0.585774	0.533065	0.052710	-0.001226	-0.001976	0.036380	0.0666	1.503158
Andre Drummond	2019	1054	562	494	0.582857	0.523322	0.059535	-0.002307	-0.002697	0.042713	0.0722	1.456962
Jayson Tatum	2022	1354	680	687	0.549133	0.495343	0.053790	-0.002530	-0.003327	0.040802	0.0770	1.399853
Kobe Bryant	2016	1118	397	516	0.444444	0.351351	0.093093	-0.006244	-0.007232	0.071320	0.0780	1.406691
Klay Thompson	2016	1384	642	692	0.514925	0.458400	0.056525	-0.003207	-0.003431	0.044638	0.0844	1.343172
T.J. Warren	2020	997	534	473	0.582857	0.525547	0.057310	-0.001594	-0.002605	0.044007	0.0858	1.361483
Mikal Bridges	2023	1267	593	618	0.520000	0.461056	0.058944	-0.002089	-0.003421	0.047302	0.0858	1.318437
Paolo Banchero	2024	1417	643	705	0.507246	0.448006	0.059240	-0.001531	-0.001889	0.045462	0.0868	1.344634
Lou Williams	2019	1140	484	561	0.489583	0.418582	0.071001	-0.002161	-0.003242	0.055088	0.0870	1.347715
Jalen Green	2024	1332	563	672	0.483516	0.418211	0.065305	-0.003935	-0.004277	0.050959	0.0898	1.365452
Bojan Bogdanovic	2021	925	406	444	0.505882	0.432143	0.073739	-0.003300	-0.004779	0.058425	0.0904	1.343922
Klay Thompson	2019	1417	662	715	0.516779	0.461356	0.055422	-0.002122	-0.002767	0.043421	0.0906	1.340115
Shai Gilgeous-Alexander	2025	1575	828	762	0.565041	0.518435	0.046606	-0.001277	-0.001396	0.036128	0.0928	1.328639
Pascal Siakam	2020	1114	507	547	0.509091	0.449203	0.059888	-0.003522	-0.004125	0.051433	0.1040	1.244565
Cade Cunningham	2022	1031	429	477	0.477778	0.410202	0.067576	-0.004642	-0.006157	0.059674	0.1042	1.235593

Observed and Permutation Statistics for Stratified Games Based on Number of Shots

Player	Season	Game ID	Number of shots	Number of makes	Number of opportunities	Number of opportunity makes	Opportunity Proportion	Number of other attempts	Other Proportion	Difference in Proportions	Null median	Null mean	Null SD	p-value	z-score	Adjusted Difference in Proportions
Tobias Harris	2017	400900394	11	6	3	2	0.66667	8	0.50000	0.16667	-0.60000	-0.279938	0.385726	0.206557	1.157828	0.446605
	2017	400899969	14	9	2	0	0.00000	12	0.75000	-0.75000	-0.16667	-0.218016	0.362861	0.995785	-1.468809	-0.531984
Krisztop Puzingis	2024	401584847	15	10	3	1	0.33333	12	0.75000	-0.41667	-0.192308	-0.177540	0.388354	0.815603	-0.706736	-0.239127
	2024	400899688	17	7	1	0	0.00000	16	0.43750	-0.43750	-0.437500	-0.239548	0.304098	0.948980	-0.650949	-0.197952
Klay Thompson	2021	401307676	19	14	5	2	0.40000	14	0.85714	-0.45714	-0.102564	-0.096319	0.242928	0.529000	-1.485314	-0.360824
Zion Williamson	2021	400899688	17	14	5	2	0.40000	16	0.85714	-0.45714	-0.102564	-0.096319	0.242928	0.529000	-1.485314	-0.360824
Joel Embiid	2022	401359889	19	8	1	0	0.00000	18	0.44444	-0.44444	-0.44444	-0.214453	0.317187	0.925795	-0.720096	-0.223991
Dellar DeRozan	2018	400975279	19	10	2	0	0.00000	17	0.588235	-0.588235	-0.229167	-0.204417	0.343287	0.995439	-1.117091	-0.383818
Jalen Williams	2025	401703417	19	8	2	1	0.50000	17	0.588235	-0.588235	-0.229167	-0.204417	0.343287	0.995439	-1.117091	-0.383818
Devin Booker	2023	401468180	20	12	2	0	0.00000	18	0.66667	-0.66667	-0.125000	-0.164589	0.326739	0.973469	-1.536633	-0.502078
Derrick Rose	2017	400899581	20	10	3	1	0.33333	17	0.529412	-0.196078	-0.196078	-0.202313	0.336399	0.50351	0.018533	0.006234
Karl-Anthony Towns	2022	401360083	21	10	2	1	0.50000	19	0.473684	0.026316	-0.166667	-0.181135	0.343550	0.441805	0.603845	0.207451
Tyler Herro	2023	401468177	21	10	2	1	0.50000	19	0.473684	0.026316	-0.166667	-0.181135	0.343550	0.441805	0.603845	0.207451
Franz Wagner	2025	401702381	22	11	6	4	0.66667	16	0.437500	0.229167	-0.192982	-0.177728	0.325467	0.102477	1.250189	0.406895
Anthony Edwards	2025	401705518	22	10	1	0	0.00000	21	0.476190	-0.476190	-0.476190	-0.191022	0.324169	0.866496	-0.879691	-0.285168
Tyrese Maxey	2024	401584107	23	9	1	0	0.00000	22	0.490909	-0.490909	-0.490909	-0.204619	0.300177	0.913600	-0.681169	-0.204472
Giannis Antetokounmpo	2019	401071616	26	15	4	1	0.25000	22	0.636364	-0.386364	-0.090909	-0.114814	0.278266	0.859013	-0.972370	-0.271550
Giannis Antetokounmpo	2021	401267680	27	15	4	3	0.75000	23	0.521739	0.226261	-0.065217	-0.146026	0.301251	0.121181	1.245225	0.374287
Jalen Brunson	2024	401585029	28	15	5	2	0.40000	23	0.565217	-0.165217	-0.041667	-0.137532	0.302072	0.791602	-0.091651	-0.027685
Shai Gilgeous-Alexander	2024	401616476	30	15	3	1	0.33333	27	0.518519	-0.185185	-0.185185	-0.152109	0.303405	0.623444	-0.109018	-0.033077
Jamal Murray	2025	401703346	36	20	3	1	0.33333	33	0.575758	-0.242424	-0.062500	-0.084695	0.247269	0.705141	-0.637883	-0.157729
Russell Westbrook	2017	400900303	36	12	3	1	0.33333	33	0.33333	0.00000	-0.342827	-0.172515	0.267189	0.155334	0.645667	0.172515
Nikola Jokic	2025	401705041	36	15	2	0	0.00000	34	0.441176	-0.441176	-0.187500	-0.151428	0.300292	0.966025	-0.964890	-0.289749
Bradley Beal	2021	401267439	37	17	2	0	0.00000	35	0.485714	-0.485714	-0.137255	-0.131316	0.294009	0.942318	-1.205400	-0.354398
Damian Lillard	2023	401468072	38	22	5	1	0.20000	33	0.636364	-0.436364	-0.088235	-0.079240	0.294151	0.933894	-1.525189	-0.357124
Dellar DeRozan	2017	400900446	38	17	5	3	0.60000	33	0.424242	0.175758	-0.123810	-0.113283	0.296108	0.19673	0.976131	0.289040
Klay Thompson	2023	401468713	39	21	4	1	0.25000	35	0.571429	-0.321429	-0.045405	-0.097487	0.250670	0.856281	-0.893329	-0.223931
Russell Westbrook	2017	400900341	39	21	6	3	0.50000	33	0.545455	-0.045455	-0.045455	-0.097487	0.250670	0.856281	-0.893329	-0.223931
Russell Westbrook	2017	400900507	40	21	3	1	0.33333	37	0.540541	-0.207207	-0.116883	-0.111033	0.250023	0.693548	-0.377122	-0.096175
Dejounte Murray	2024	401585688	44	18	3	0	0.00000	41	0.439024	-0.439024	-0.081301	-0.134811	0.276355	0.996681	-1.100807	-0.304213